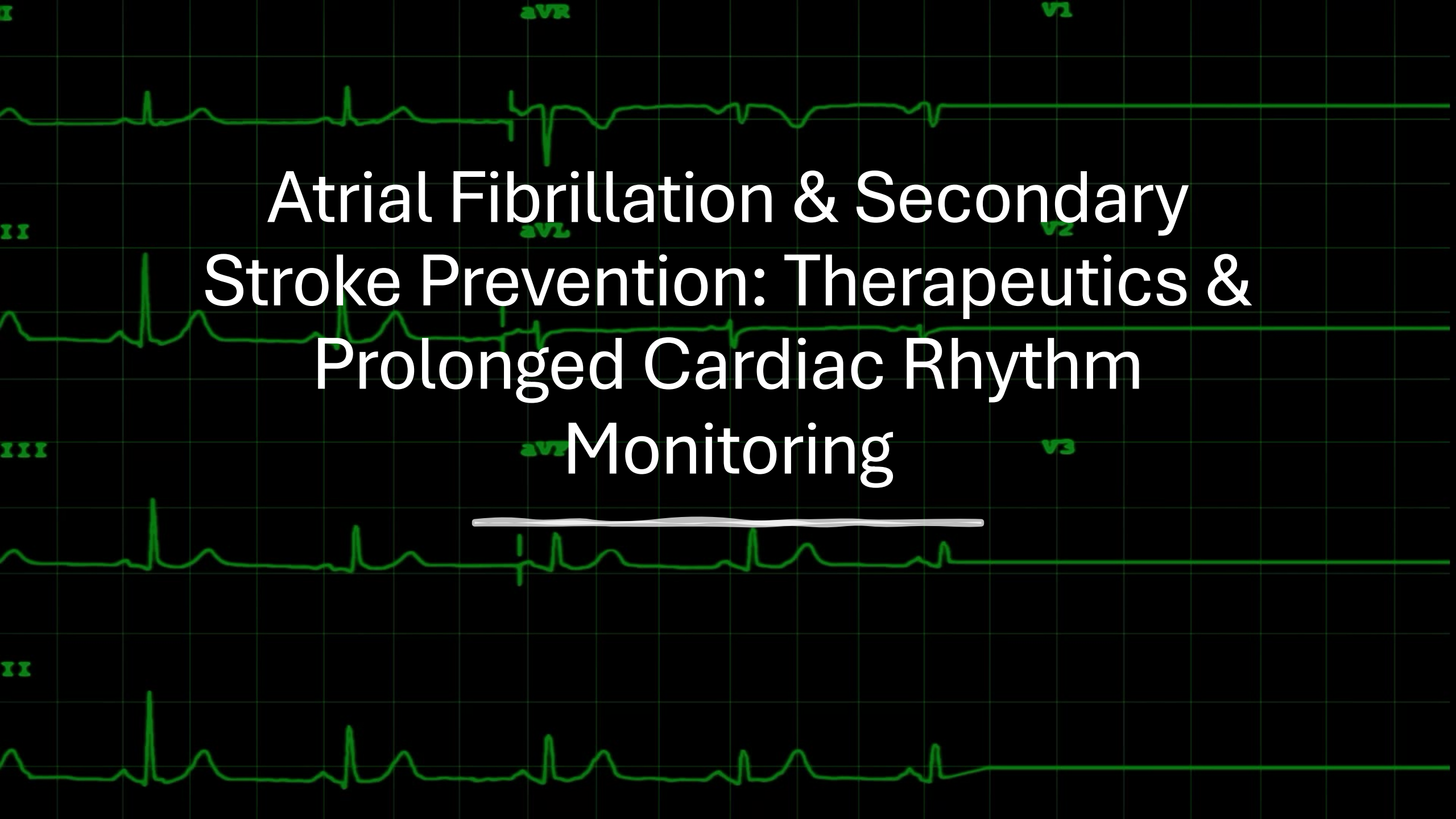




Atrial Fibrillation & Secondary Stroke Prevention: Therapeutics & Prolonged Cardiac Rhythm Monitoring



Objectives

- Atrial fibrillation (AF) and stroke risk
- Anticoagulation (AC)
 - Special populations
 - Timing of initiation after stroke
- Ischemic stroke (IS) despite AC
- Rhythm control & Factor XI/XIa inhibitors
- AF evaluation after IS
 - Prolonged cardiac rhythm monitoring
 - AF burden and IS risk
 - Atrial high-rate episodes
 - Wearables

Atrial Fibrillation

- Global prevalence: ~59.7 million (2019)
- Lifetime risk ~33% (US, ≥ 45 y)
 - $\uparrow$ Caucasians
 - $M > F$

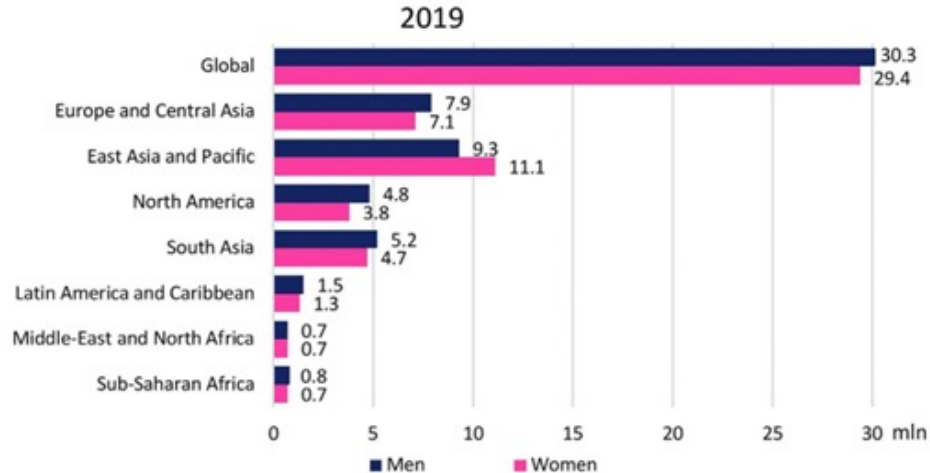


Fig. 2: Prevalence of atrial fibrillation/atrial flutter in both men and women.

Lancet Reg Health Eur. 2024 Feb 1;37:100786.

BMC Public Health. 2022 Nov 3;22(1):2015.

J Am Coll Cardiol. 2020 Apr 14;75(14):1689-1713.

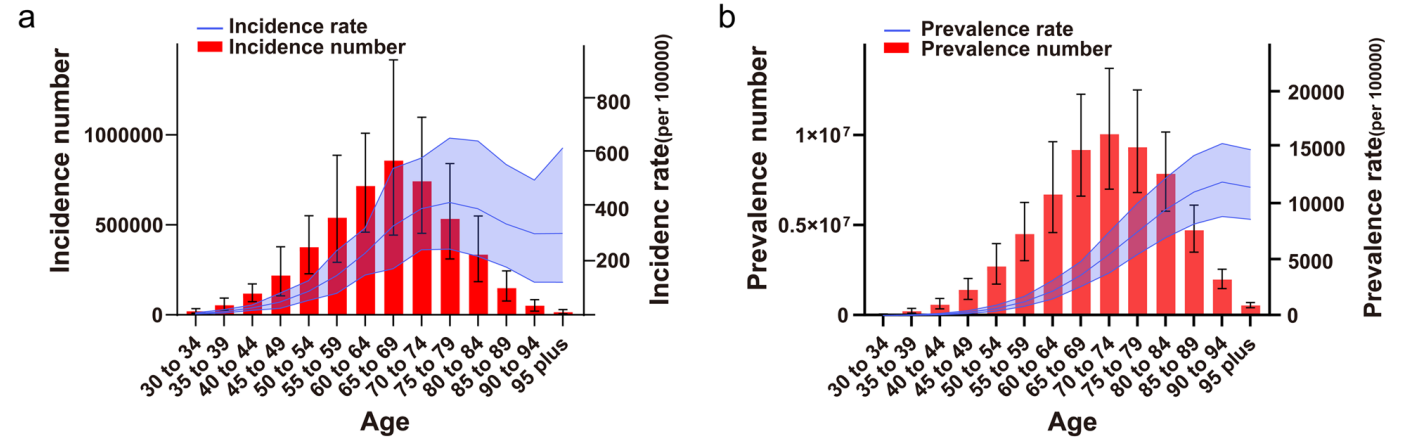
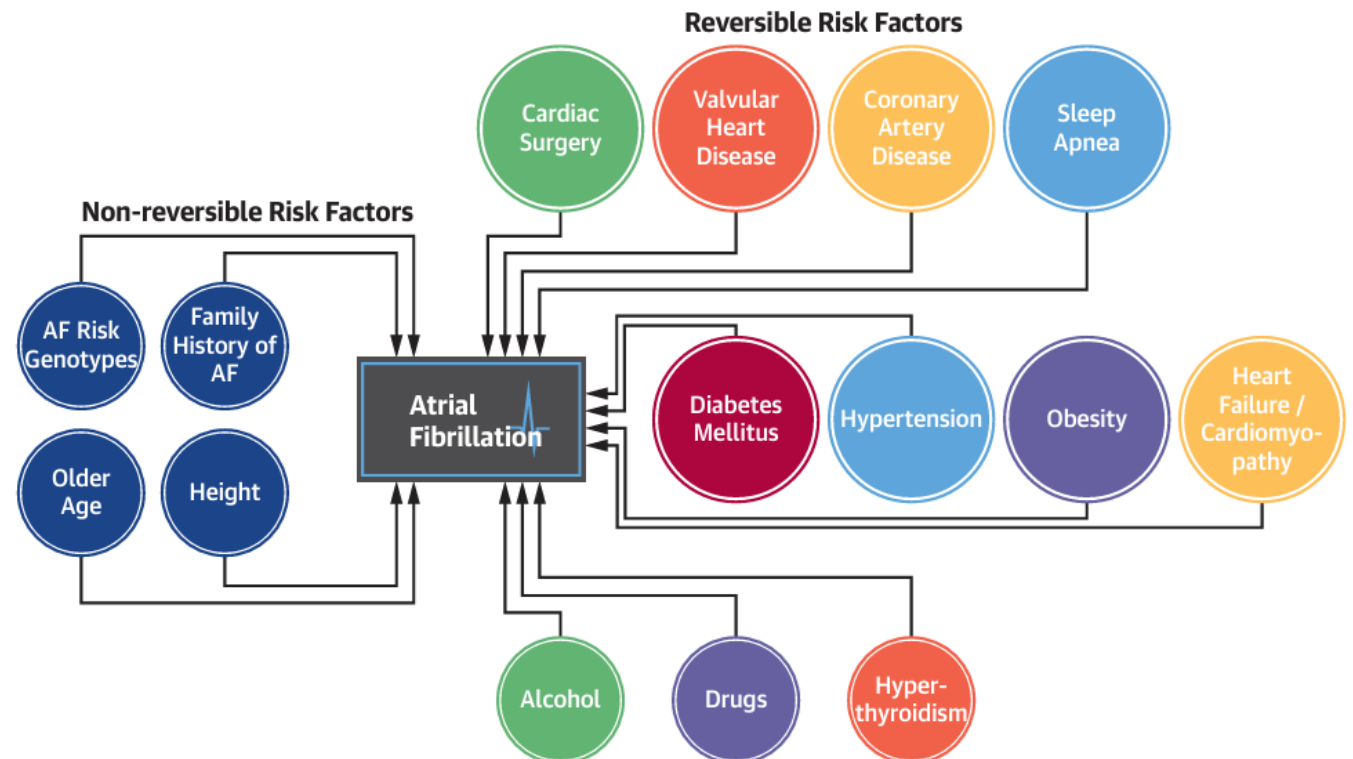


FIGURE 2 Potential Modifiable and Nonmodifiable Risk Factors for AF



AF & Stroke Risk

- Non-valvular AF: ↑ 5x risk
 - AF w/ mitral stenosis: ↑ 20x
- 20-30% of IS
- Risk prediction models
 - CHA₂DS₂-VASc
 - * female sex → risk modifier
 - ATRIA, GARFIELD
 - ML → c-index ~0.87

*multimorbidity

*risk is dynamic

J Am Coll Cardiol. 2020 Apr 14;75(14):1689-1713.

Thromb Haemost. 2022 Jan;122(1):142-150.

Thromb Haemost. 2019 Jul;119(7):1162-1170.

2026 repairs 15 | linked source notes

TABLE 4 CHA₂DS₂-VASc Score for Prediction of Stroke in Atrial Fibrillation Patients

C	Congestive heart failure	1
H	Hypertension, including treated / controlled	1
A	Age ≥75 yrs	2
D	Diabetes mellitus	1
S ₂	Prior TIA or stroke	2
V	Vascular disease (MI, aortic plaque, and so on)	1
A	Age 65-74 yrs	1
Sc	Sex category (female = 1 point)	1

MI = myocardial infarction; TIA = transient ischemic attack.

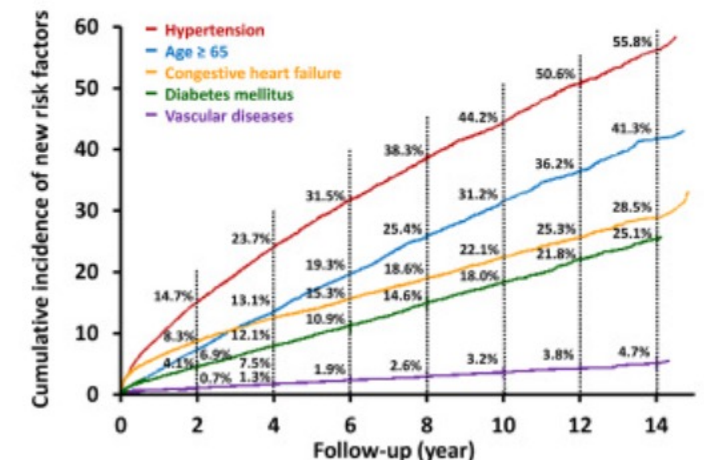
Calibration matters

Absolute risk is cohort-dependent.

Original bar values have not been traced to a primary cohort; do not use them as an individual risk estimate.

See Repair 15.

Cumulative incidence of new stroke risk factors for patients with a baseline CHA₂DS₂-VASc score of 0 (M) or 1 (F)



Thromboembolism

- **Stasis**
- **Atrial structure (remodeling, dilation) & contraction**
- **Hypercoagulability**
- **Endothelial dysfunction**

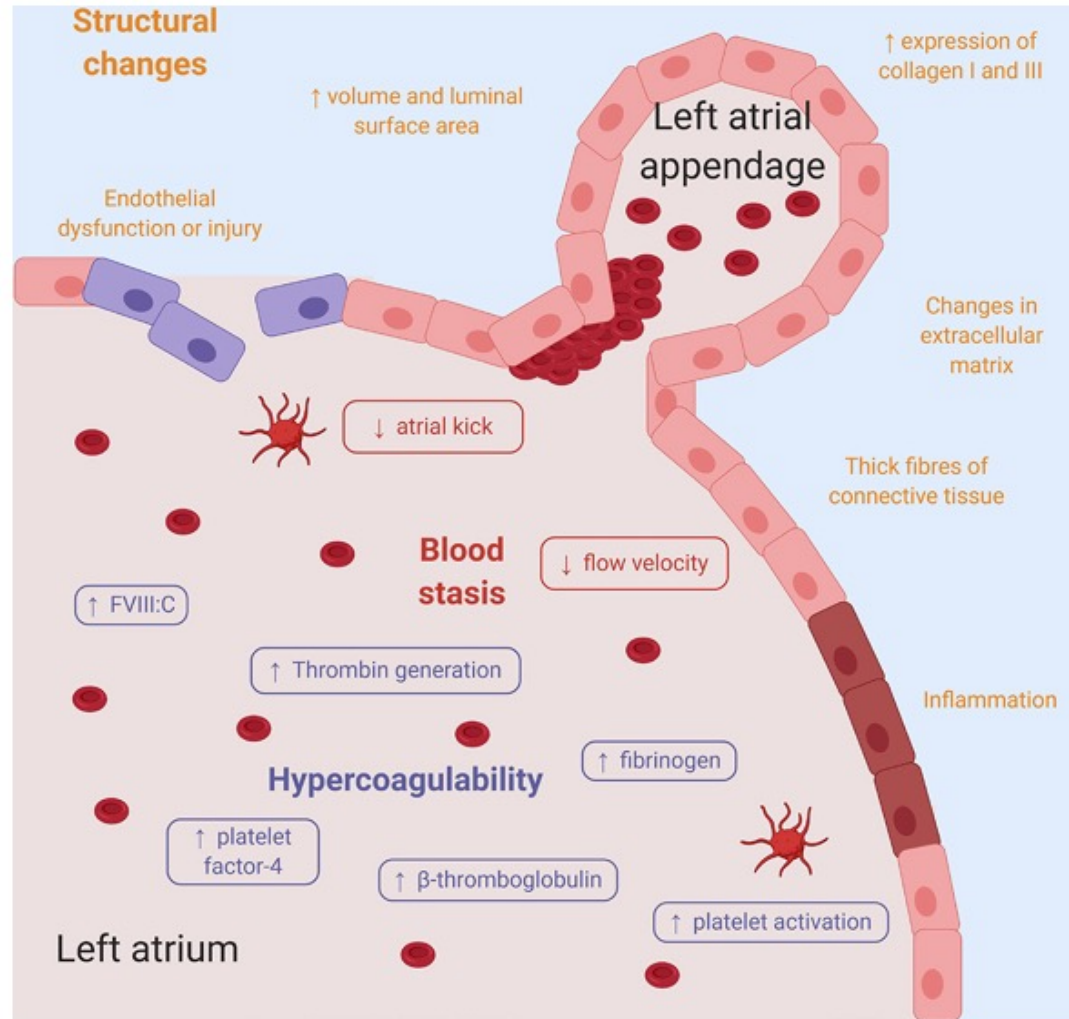


Figure 1 Effects of atrial fibrillation on the various elements of Virchow's triad (hypercoagulability, structural changes and blood stasis).

AF Management

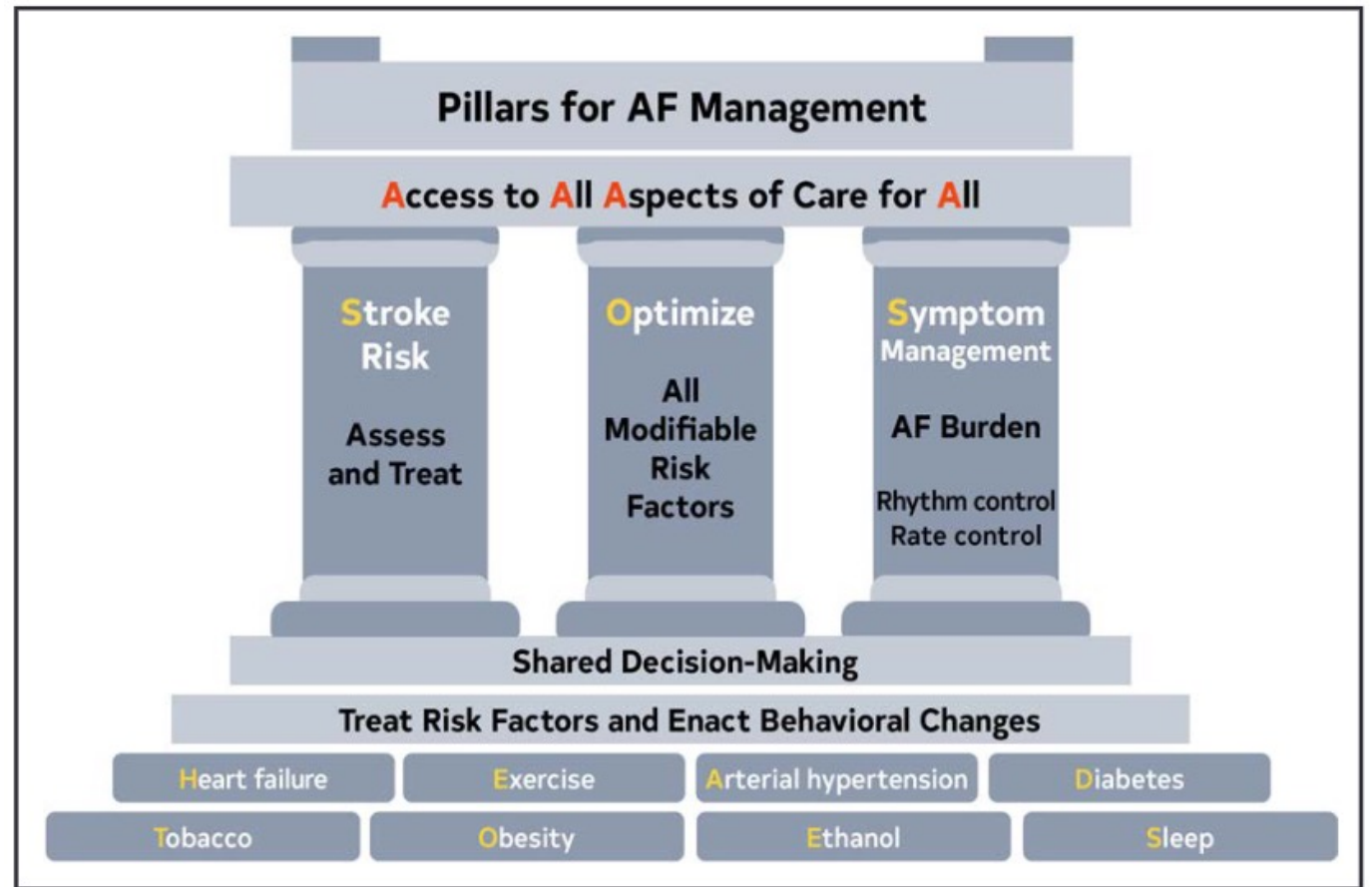
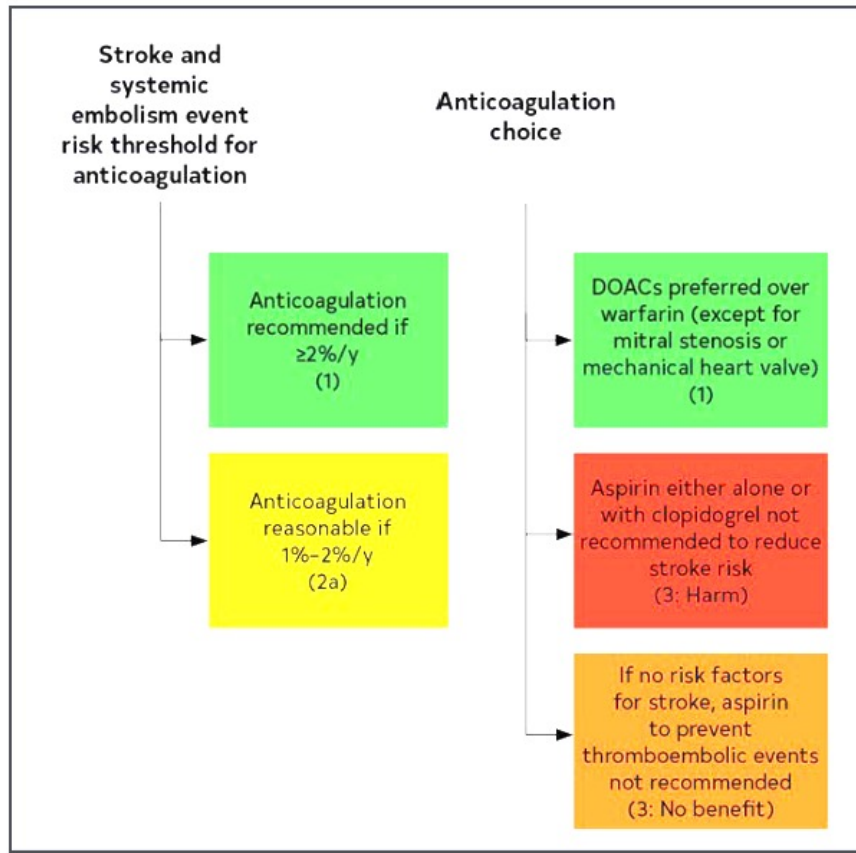


Figure 10. Antithrombotic Options in Patients With AF.

Antiplatelets & Warfarin

- Aspirin → RRR ~22%
* SPAF-1
- Aspirin + Clopidogrel
 - ACTIVE A → RR 0.72
 - ARR: 0.9%
 - Major hemorrhage (DAPT): ~2%/yr
 - ACTIVE W → RR 0.69
 - Warfarin vs aspirin/clopidogrel
- Warfarin → RRR ~64%

B Stroke

Chest. 2018 Nov;154(5):1121-1201.
Ann Intern Med. 2007 Jun 19;146(12):857-67.
N Engl J Med. 2009 May 14;360(20):2066-78.
Lancet. 2006 Jun 10;367(9526):1903-12.
Stroke. 2006 Feb;37(2):447-51.

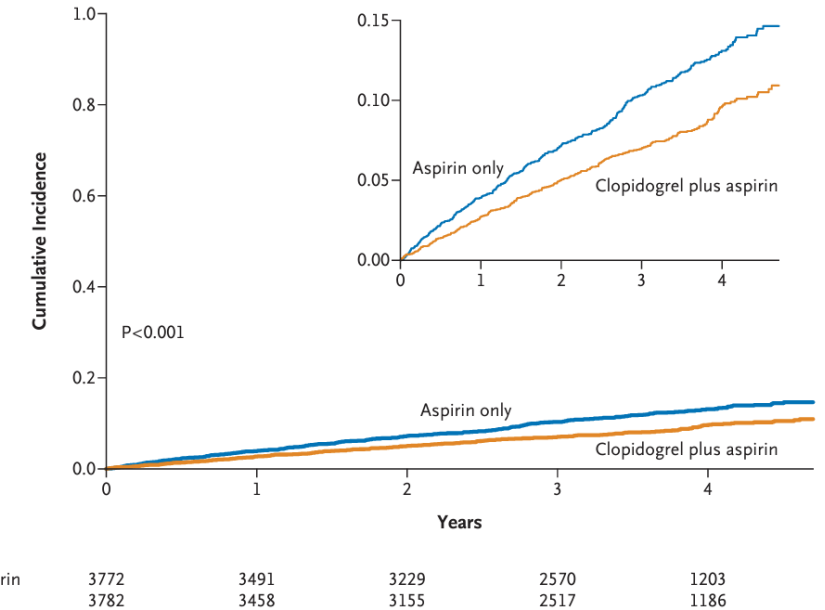


Table 6. Meta-analysis of the Efficacy of Antithrombotic Therapies for Stroke Prevention in Patients Who Have Atrial Fibrillation*

Comparison	Trials, <i>n</i>	Patients, <i>n</i>	Strokes, <i>n</i>	Relative Risk Reduction, %	Hypothetical NNT: Primary Prevention†	Hypothetical NNT: Secondary Prevention†
Adjusted-dose warfarin versus control	6	2900	186	64	40	14
Antiplatelet agents versus control	8	4876	488	22	111	34
Adjusted-dose warfarin versus antiplatelet agents	12	12 963	546	39	81	24

Direct Oral Anticoagulants – Efficacy & Safety

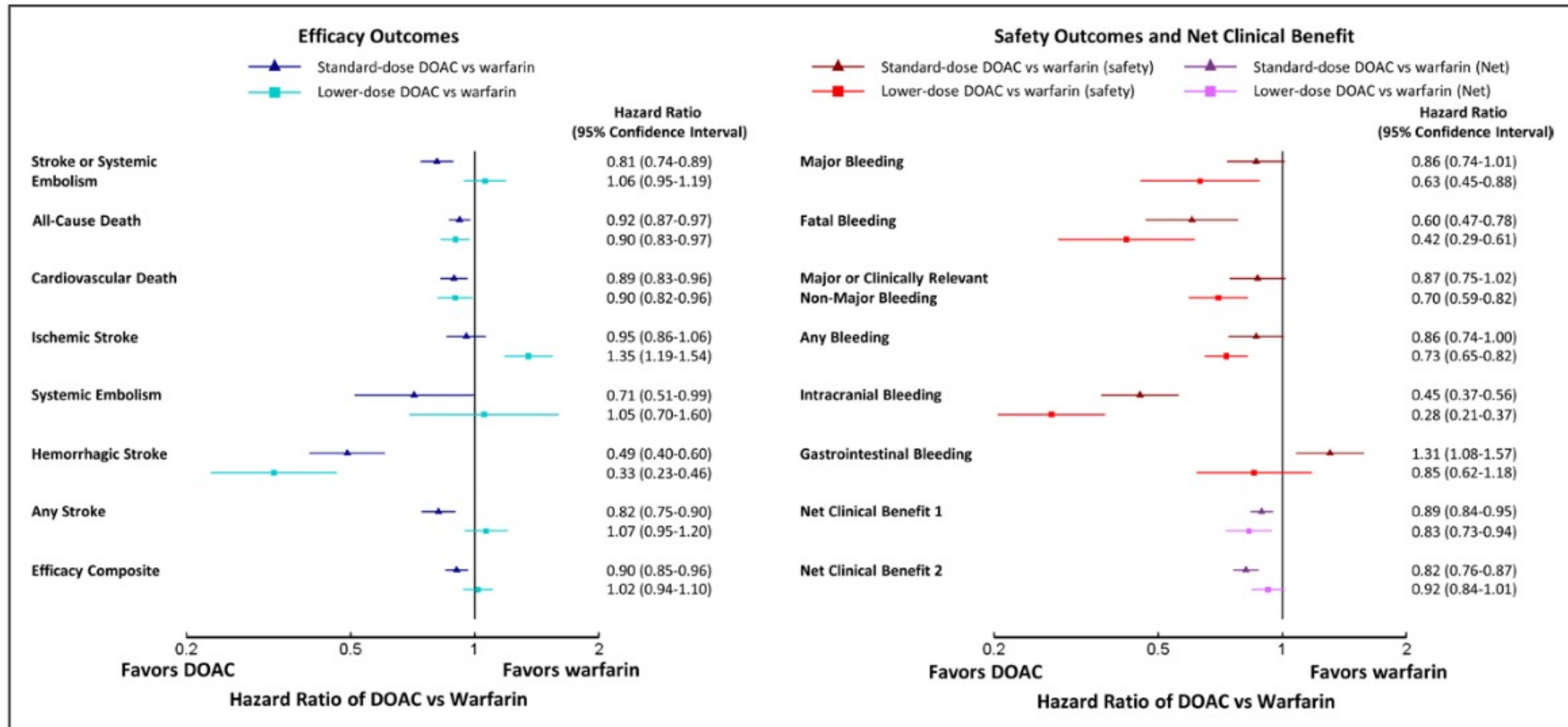


Figure 1. Comparison of standard-dose and lower-dose direct oral anticoagulants vs warfarin for efficacy, safety, and net clinical benefit outcomes.

AVERROES

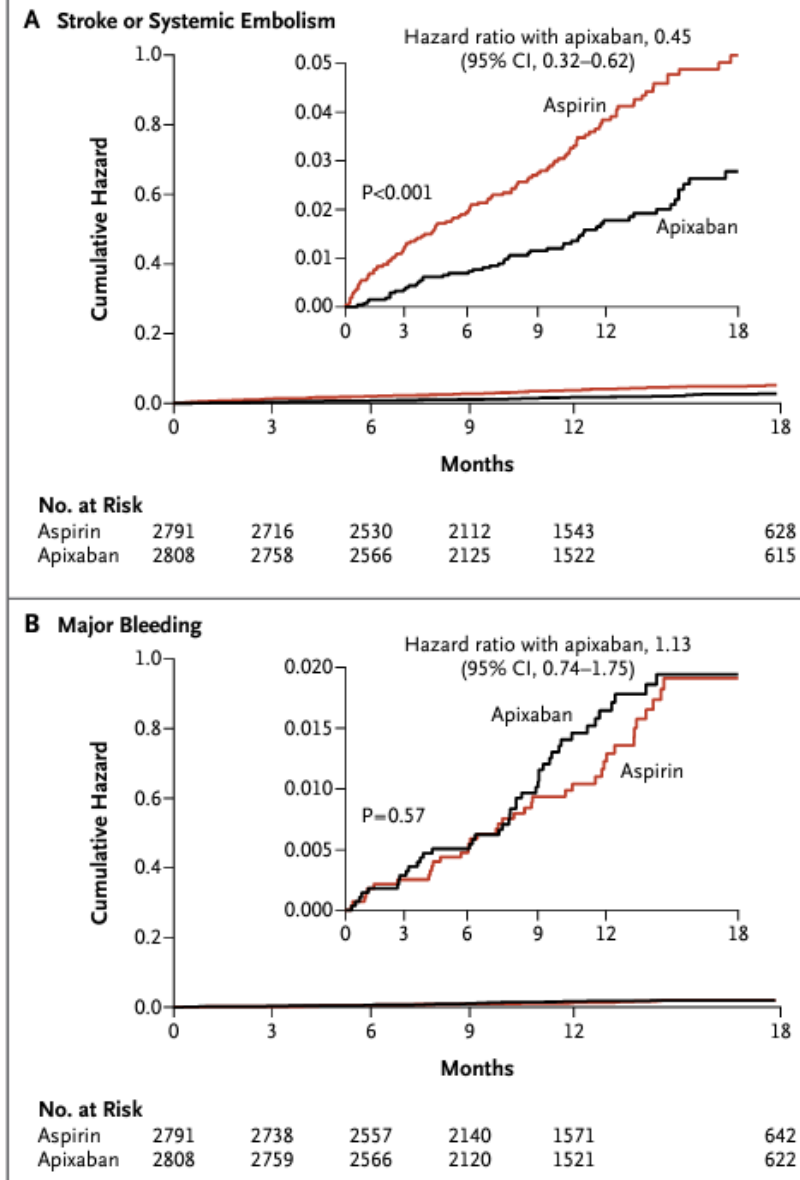


Figure 1. Cumulative Hazard Rates for the Primary Efficacy and Safety Outcomes, According to Treatment Group.

Effectiveness

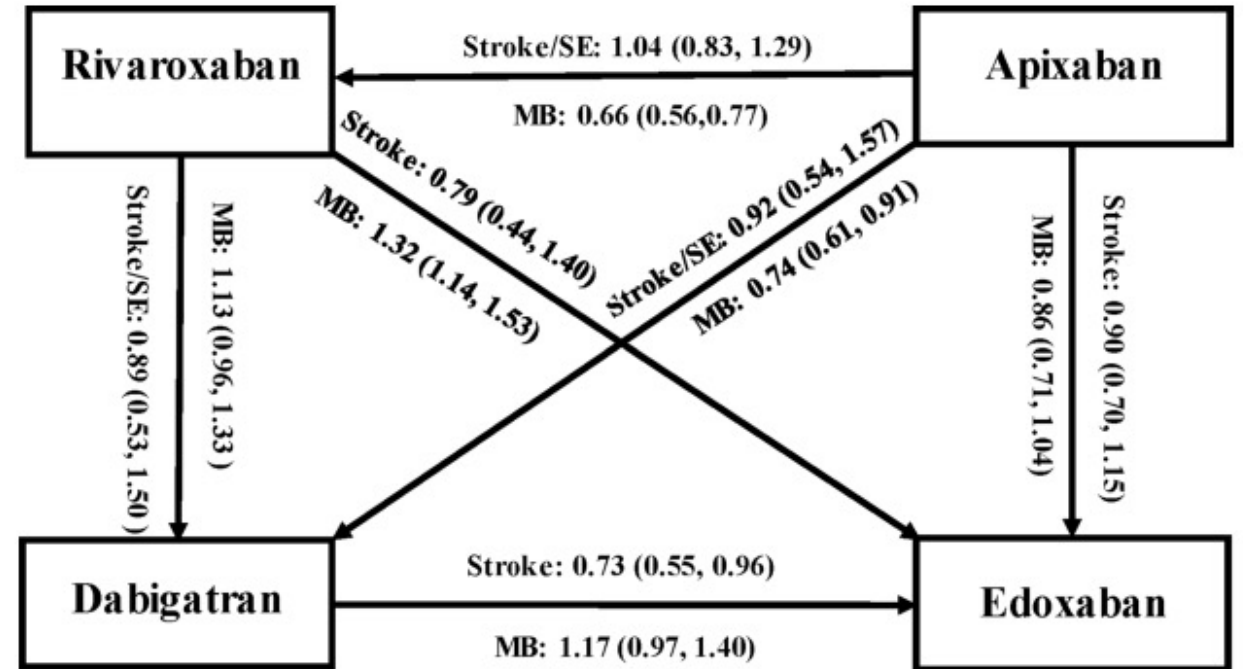


Fig. 1 Indirect comparative effectiveness and safety amongst DOACs in patients with AF

Special Populations

- DOAC trials for AF – exclusions:
 - pregnancy
 - **mechanical heart valve**
 - moderate-severe MV stenosis
 - ESRD/dialysis



N Engl J Med. 2013 Sep 26;369(13):1206-14.
 NEJM Evid. 2023 Jul;2(7):EVIDoa2300067.
 Ann Biomed Eng. 2015 Apr;43(4):844-57.

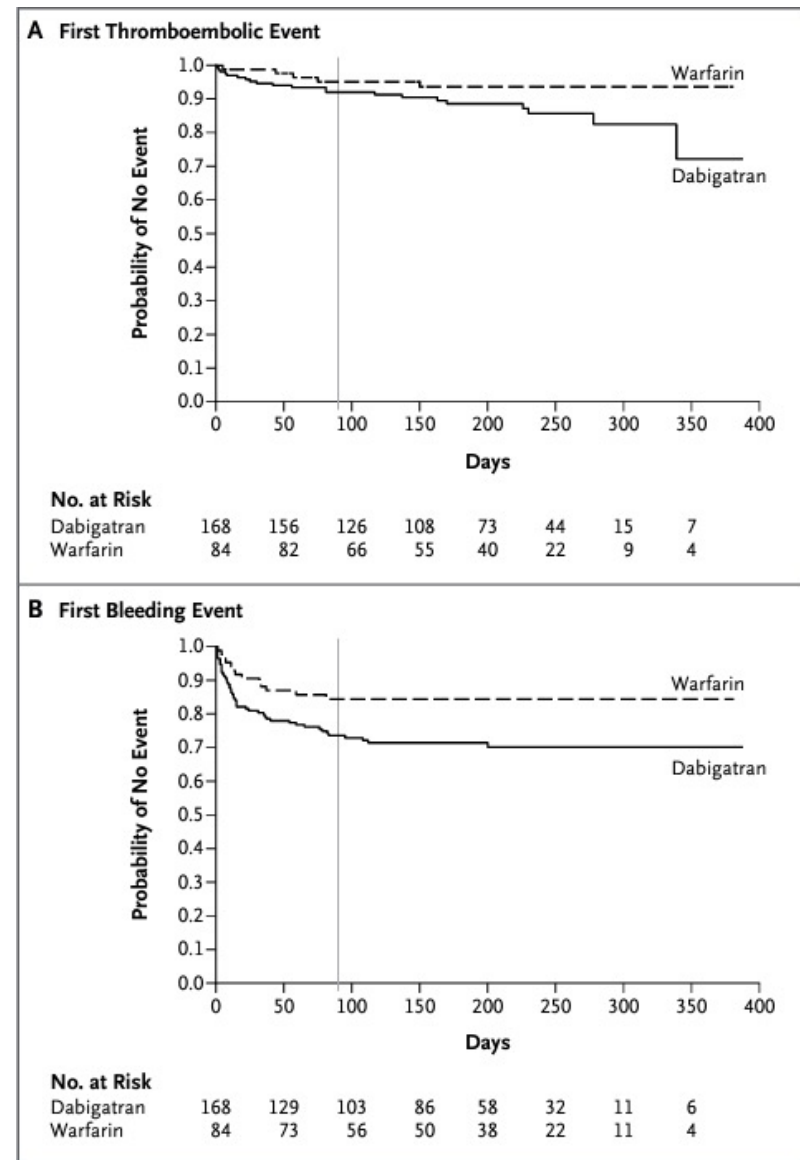


Figure 1. Kaplan–Meier Analysis of Event-free Survival.

Panel A shows event-free survival from the first thromboembolic event (i.e., stroke, systemic embolism, transient ischemic attack, or myocardial infarction) or death ($P=0.24$). Panel B shows event-free survival from the first bleeding event ($P=0.01$). In each panel, the vertical line indicates the start of the RE-ALIGN extension trial (RE-ALIGN-EX) and the P value was calculated with the use of the Wald chi-square test.



On-X Life Technologies

PROACT Xa

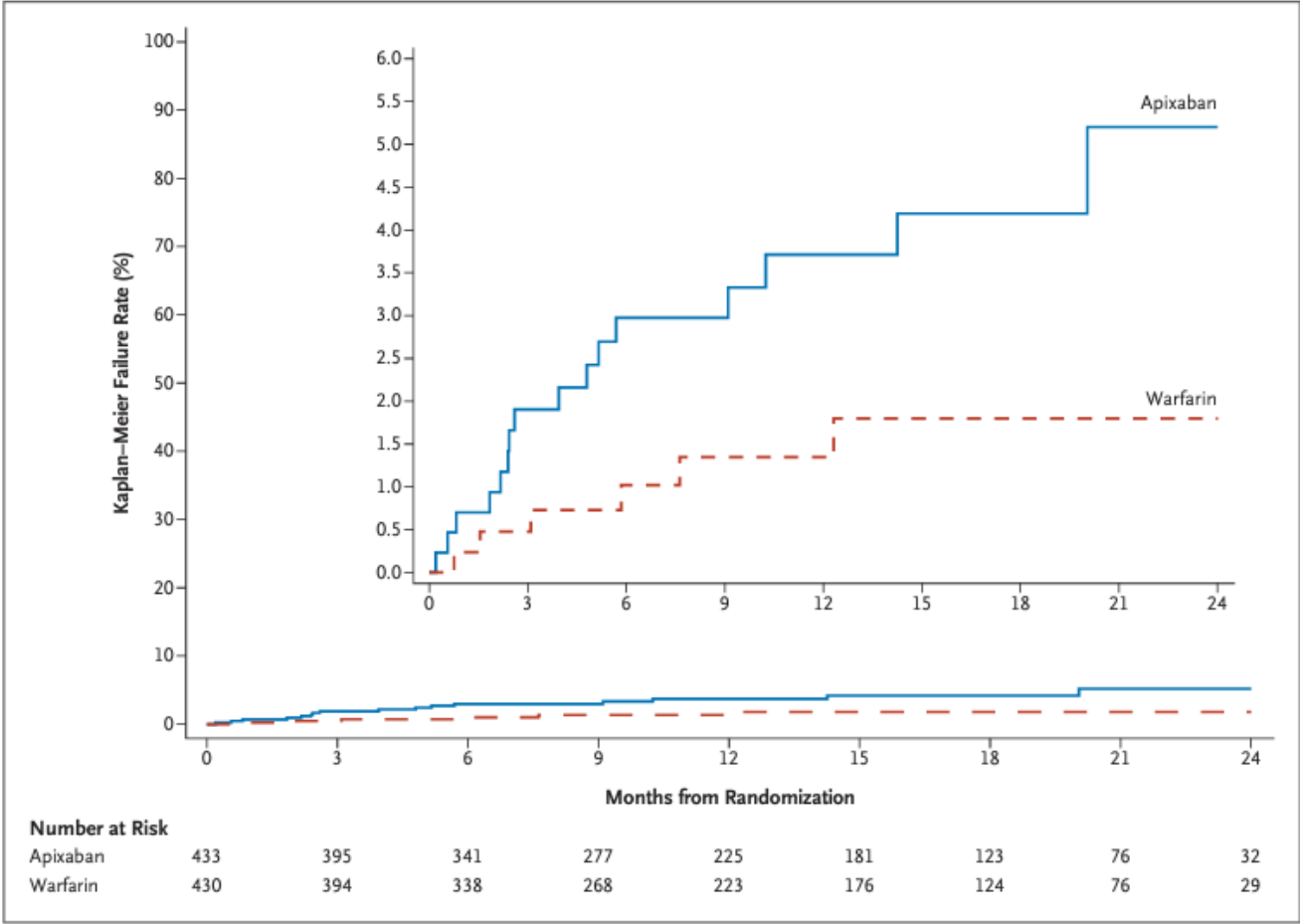
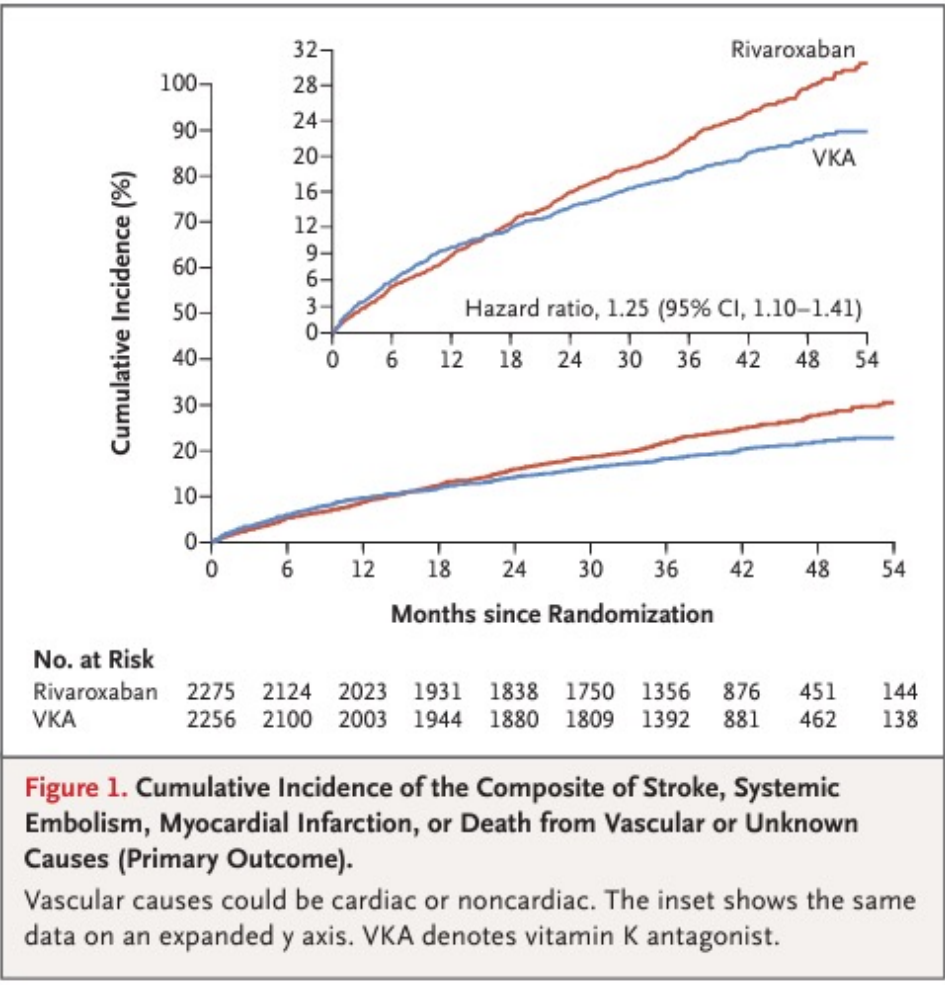


Figure 2. Cumulative Incidence of Valve Thrombosis or Valve-Related Thromboembolism.

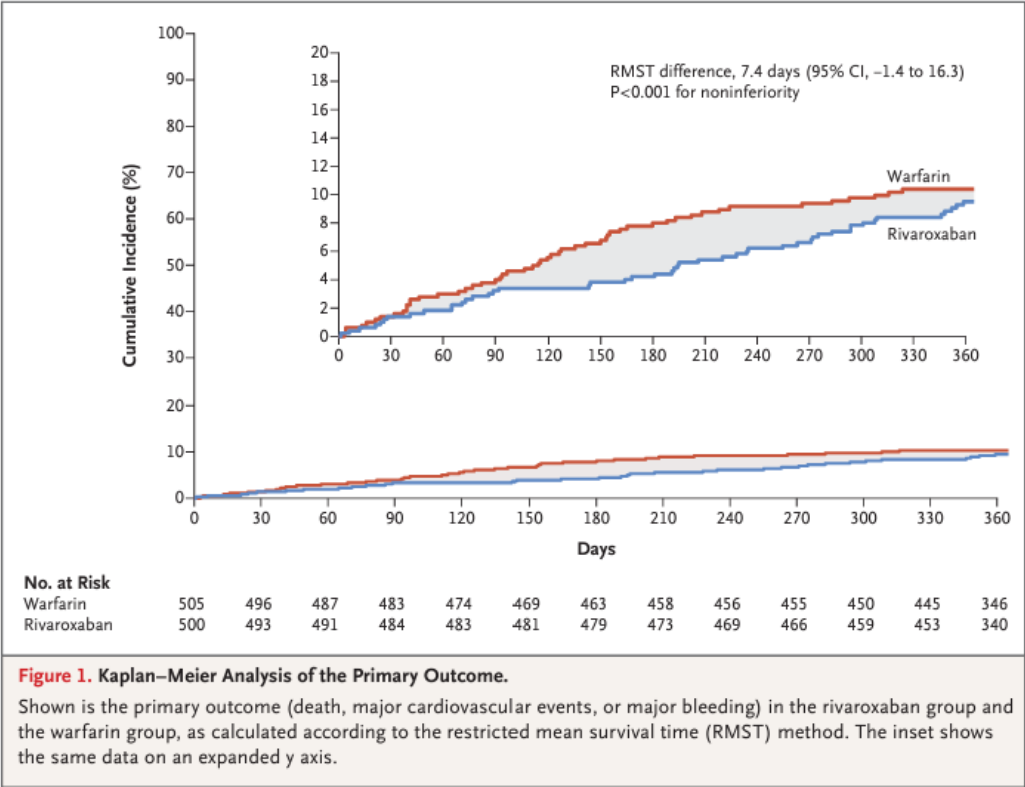
Mitral Valve Stenosis

INVICTUS

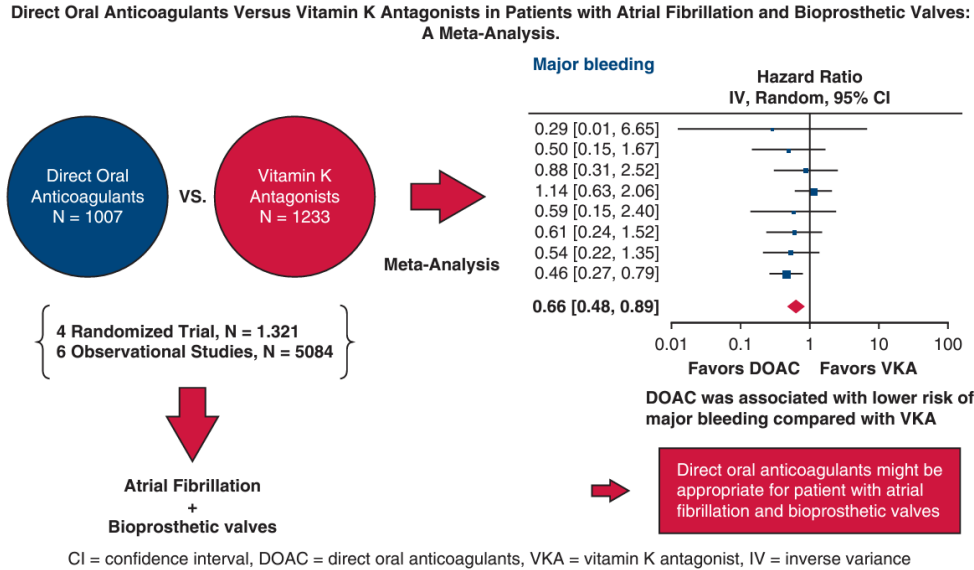
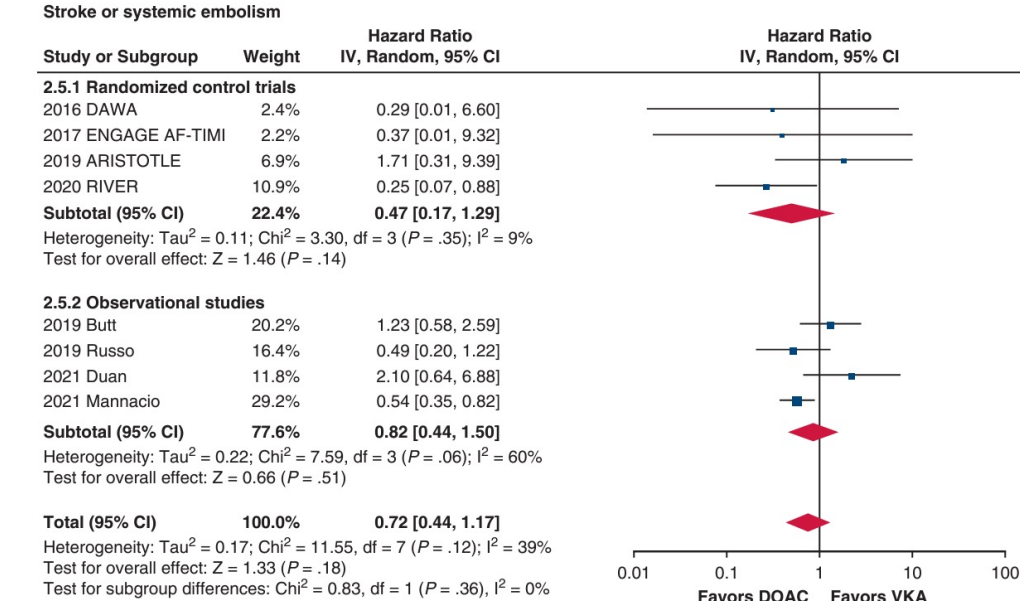


Bioprosthetic Valves

RIVER



TAVI + AF: ENVISAGE had more major bleeding with edoxaban.
ATLANTIS did not establish apixaban equivalence. See Repair 12.



ESRD / Dialysis

- CKD → risk factor for stroke in AF
- Stage IV CKD (observational) → DOAC

→ apixaban or warfarin reasonable in ESRD

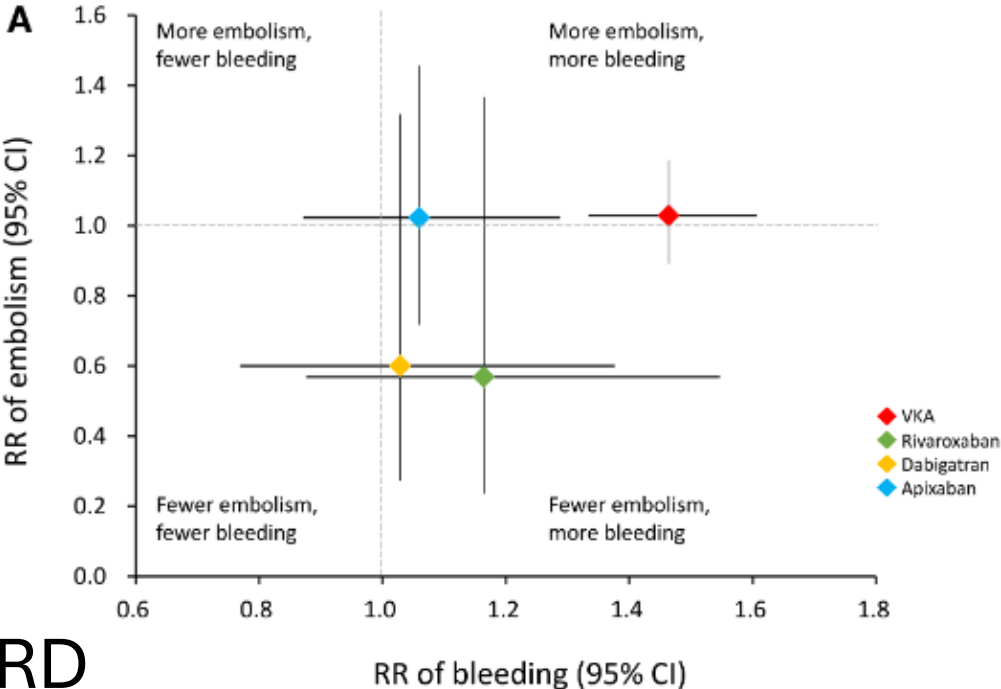


Table 19. Recommended Doses of Currently Approved DOACs According to Renal Function

DOAC	CrCl (mL/min)				
	>95	51-95	31-50	15-30	<15 or on dialysis
Apixaban	5 or 2.5 mg twice daily*	5 or 2.5 mg twice daily*	5 or 2.5 mg twice daily*	5 or 2.5 mg twice daily*	5 or 2.5 mg twice daily*
Dabigatran	150 mg twice daily	150 mg twice daily	150 mg twice daily	75 mg twice daily	Contraindicated
Edoxaban	Contraindicated	60 mg once daily	30 mg once daily	30 mg once daily	Contraindicated
Rivaroxaban	20 mg once daily	20 mg once daily	15 mg once daily	15 mg once daily	15 mg once daily†

Table 19 must be read with the restored dosing/evidence footnotes in Repair 09 (linked appendix).

Circulation. 2024 Jan 2;149(1):e167.

J Am Heart Assoc. 2024 Apr 16;13(8):e034176.

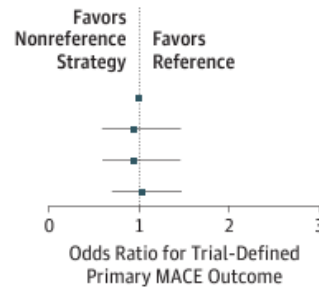
Percutaneous Coronary Intervention

Figure 3. Forest Plots for Efficacy Outcomes

A Trial-defined primary MACE

Odds ratio (95% CI)

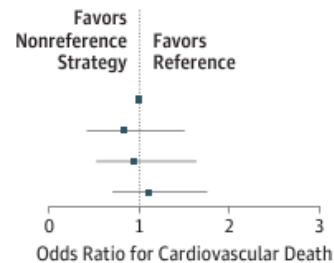
VKA + DAPT (reference)	
VKA + P2Y ₁₂ inhibitor	0.96 (0.60-1.46)
NOAC + DAPT	0.94 (0.60-1.15)
NOAC + P2Y ₁₂ inhibitor	1.02 (0.71-1.97)



C Cardiovascular death

Odds ratio (95% CI)

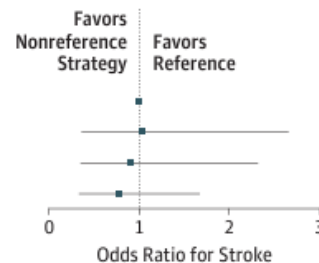
VKA + DAPT (reference)	
VKA + P2Y ₁₂ inhibitor	0.82 (0.42-1.49)
NOAC + DAPT	0.94 (0.53-1.63)
NOAC + P2Y ₁₂ inhibitor	1.11 (0.70-1.75)



E Stroke

Odds ratio (95% CI)

VKA + DAPT (reference)	
VKA + P2Y ₁₂ inhibitor	1.02 (0.36-2.65)
NOAC + DAPT	0.91 (0.35-2.32)
NOAC + P2Y ₁₂ inhibitor	0.77 (0.34-1.67)



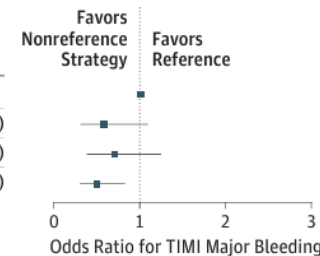
- DOAC + P2Y₁₂ inhibitor: ↓ risk of major hemorrhage & ICH

Figure 2. Forest Plots for Safety Outcomes

A TIMI major bleeding

Odds ratio (95% CI)

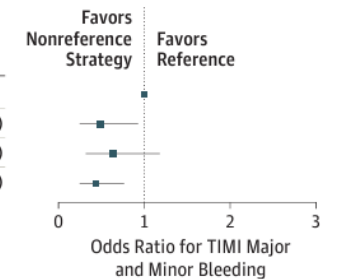
VKA + DAPT (reference)	
VKA + P2Y ₁₂ inhibitor	0.58 (0.31-1.08)
NOAC + DAPT	0.70 (0.38-1.23)
NOAC + P2Y ₁₂ inhibitor	0.49 (0.30-0.82)



B TIMI major and minor bleeding

Odds ratio (95% CI)

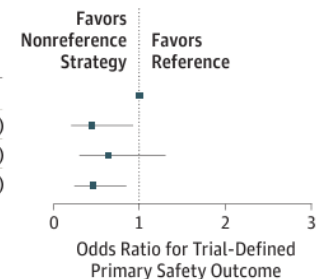
VKA + DAPT (reference)	
VKA + P2Y ₁₂ inhibitor	0.49 (0.26-0.92)
NOAC + DAPT	0.63 (0.33-1.17)
NOAC + P2Y ₁₂ inhibitor	0.43 (0.25-0.76)



C Trial-defined primary safety outcome

Odds ratio (95% CI)

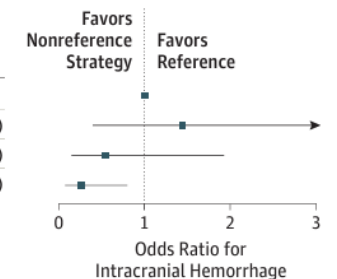
VKA + DAPT (reference)	
VKA + P2Y ₁₂ inhibitor	0.45 (0.21-0.92)
NOAC + DAPT	0.64 (0.31-1.31)
NOAC + P2Y ₁₂ inhibitor	0.47 (0.25-0.85)



D Intracranial hemorrhage

Odds ratio (95% CI)

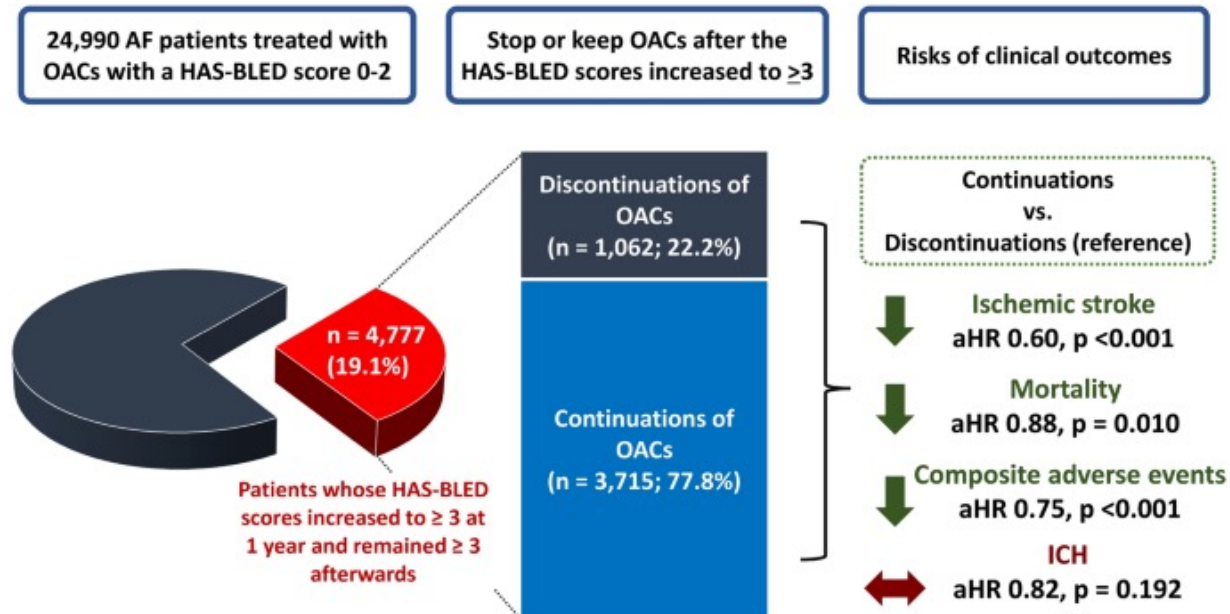
VKA + DAPT (reference)	
VKA + P2Y ₁₂ inhibitor	1.44 (0.40-5.22)
NOAC + DAPT	0.54 (0.15-1.92)
NOAC + P2Y ₁₂ inhibitor	0.26 (0.08-0.79)



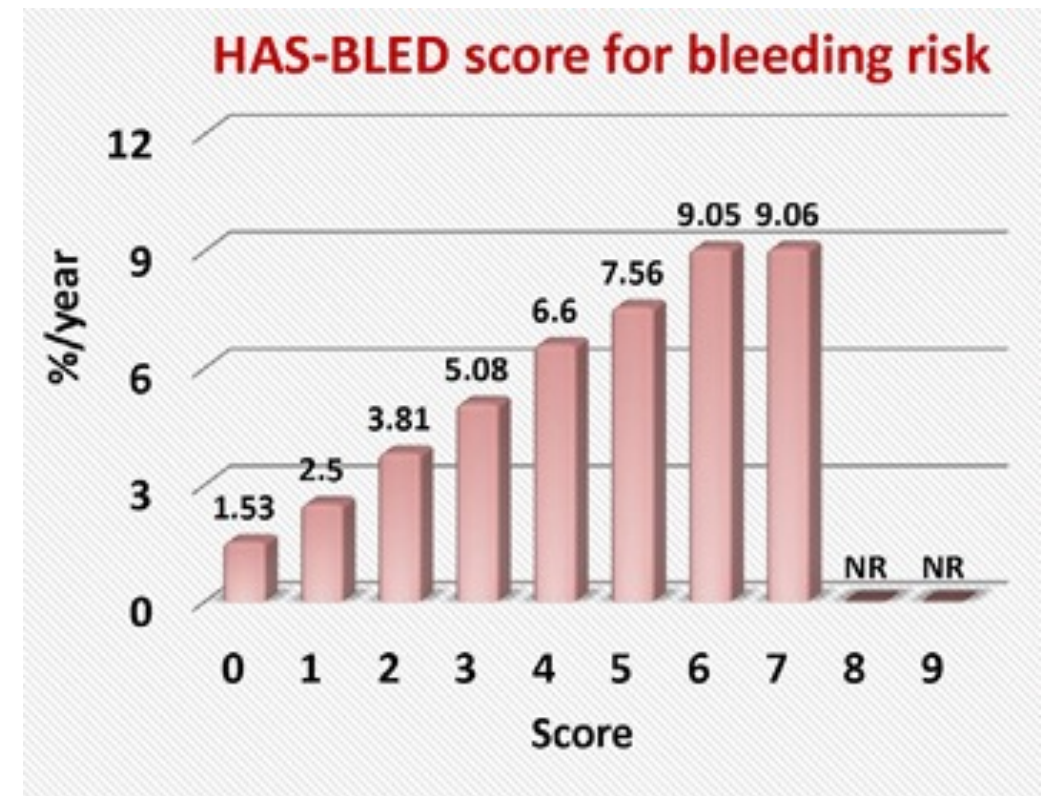
Bleeding Risk $\neq$ Avoid Anticoagulation

- Identify & correct modifiable risk factors
- HAS-BLED $\rightarrow$ c-index $\sim 0.6-0.7$
 - CHA₂DS₂-VASc=1-2 & HAS-BLED $\geq 3 \rightarrow$ OAC confers lower risk of ischemic stroke + ICH+ mortality

* risk is dynamic



Condition	Points
H – Hypertension	1
A – Ab(N) liver/renal	1 point each
S – Stroke	1
B – Bleeding	1
L – Labile INRs	1
E – Elderly (>65)	1
D – Drugs or ETOH	1 point each



Lancet Reg Health Eur. 2024 Feb 1;37:100797.
 Clin Res Cardiol. 2022 Jan;111(1):23-33.
 Am J Med. 2020 Oct;133(10):1195-1202.e2.

Timing of AC After IS

	TIMING (NCT02961348) ⁵⁴	ELAN (NCT03148457) ^{16,55,56}	OPTIMAS (NCT03759938) ⁵⁷	START (NCT03021928)
Status	Completed	Completed	Recruitment completed, awaiting results	Recruitment completed, results presented at the International Stroke Conference 2024 (Feb 7, 2024, Phoenix, AZ, USA), not yet published
Sample size	888 participants (planned 3000)	2013 participants	3648 participants	200 of 1500 planned participants (1000 patients with mild or moderate stroke and 500 with severe stroke)
Early start group	≤4 days after acute ischaemic stroke	<48 h after symptom onset (minor and moderate stroke) or at day 6 (±1 day) after symptom onset (major stroke)	≤4 days after acute ischaemic stroke	Time-to-treatment delay of 3, 6, 10, or 14 days for mild or moderate stroke; 6, 10, 14, or 21 days for severe stroke
Late start group	5–10 days after acute ischaemic stroke	Current recommendations (ie, minor stroke after day 3 [±1 day], moderate stroke after day 6 [±1 day], and major stroke after day 12 [±2 days])	7–14 days after acute ischaemic stroke	Time-to-treatment delay of 3, 6, 10, or 14 days for mild or moderate stroke; 6, 10, 14, or 21 days for severe stroke
Primary outcome	Composite outcome (recurrent ischaemic stroke, symptomatic intracerebral haemorrhage, or all-cause mortality)	Composite outcome (major bleeding, recurrent ischaemic stroke, systemic embolism, or vascular death)	Composite outcome at 90 days (recurrent symptomatic ischaemic stroke, symptomatic intracranial haemorrhage [including extradural, subdural, subarachnoid, and intracerebral haemorrhage and haemorrhagic transformation of the qualifying infarct], and systemic embolism)	Composite of any CNS haemorrhagic or other major haemorrhagic events and the ischaemic events of stroke or systemic embolism within 30 days of the index stroke
Time of assessment	90 days	30 days	90 days	30 days

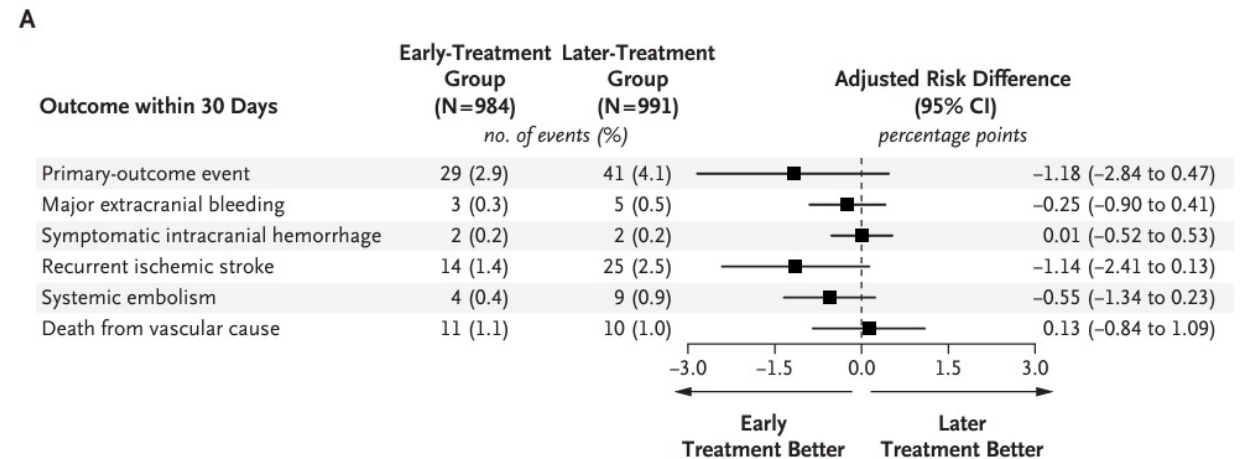
- Small-moderate infarct → ≤48 hours
- Large infarct → 1-2 weeks

Early vs Later AC for Stroke w/ AF (ELAN)

- *Population*: acute IS + AF
- *Intervention/Control*: early/late DOAC
 - Early: ≤48h minor/moderate, 6/7d major
 - Late: 3/4d minor, 6/7d moderate, 12/13/14d major
- *1° Outcome*: composite – recurrent IS, SE, sx ICH, major extracranial hemorrhage, vascular death @ 30d

Admission NIHSS: median 5 (IQR 2-11).

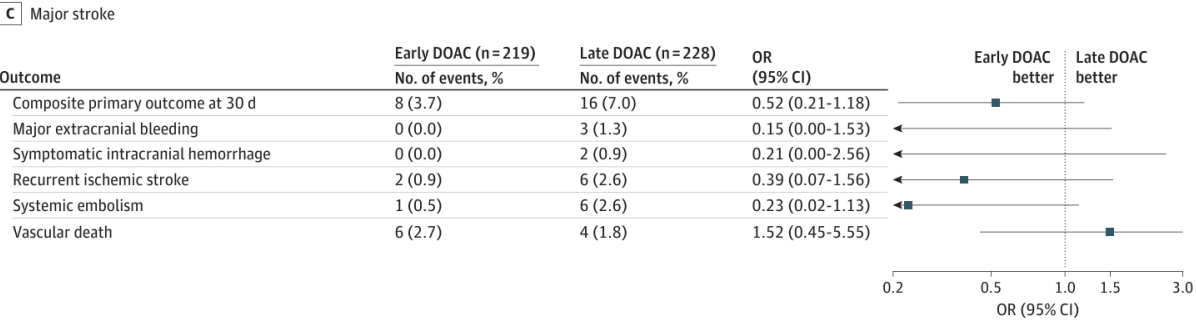
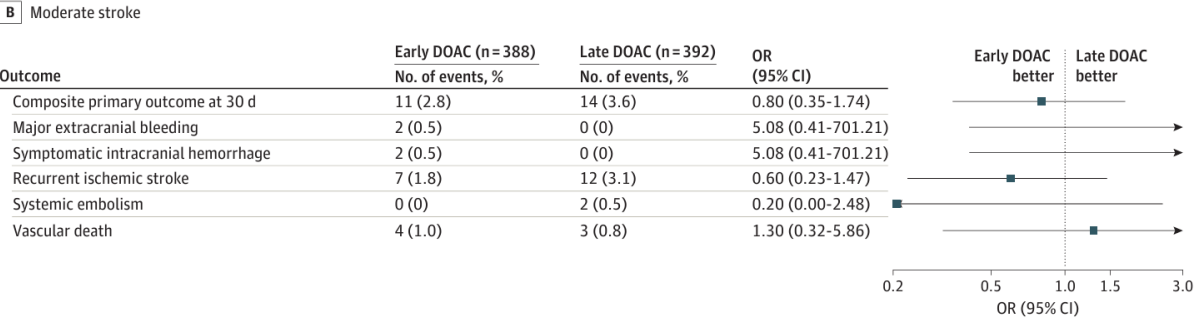
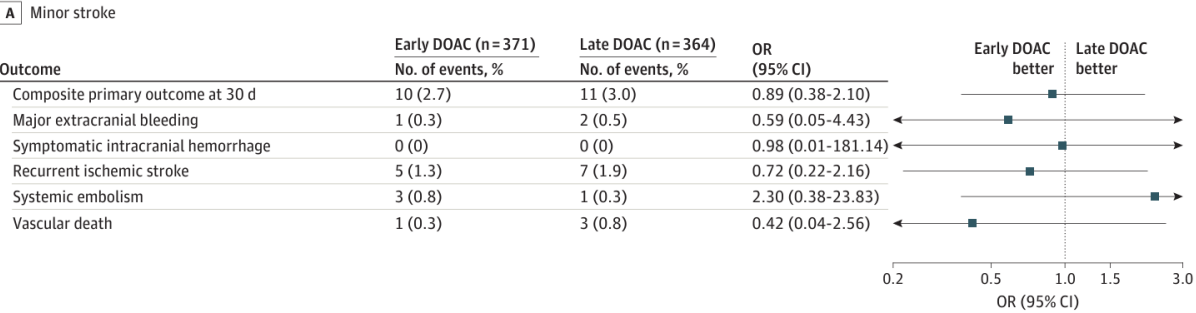
- Minor (37%): ≤1.5cm
- Moderate (40%): cortical superficial branch of MCA/ACA/PCA distribution
- Major (23%): large volume in MCA/ACA/PCA distribution OR >1.5cm in brainstem/cerebellum



Outcome	Early-Treatment Group (N = 1006) no./total no. (%)	Later-Treatment Group (N = 1007) no./total no. (%)	Adjusted Odds Ratio (95% CI)*
Primary outcome: composite outcome at 30 days	29/1006 (2.9)†	41/1007 (4.1)†	0.70 (0.44 to 1.14)‡
Secondary outcomes at 30 days			
Major extracranial bleeding	3/984 (0.3)	5/991 (0.5)	0.63 (0.15 to 2.38)
Symptomatic intracranial hemorrhage	2/984 (0.2)	2/991 (0.2)	1.02 (0.16 to 6.59)
Recurrent ischemic stroke	14/984 (1.4)	25/991 (2.5)	0.57 (0.29 to 1.07)

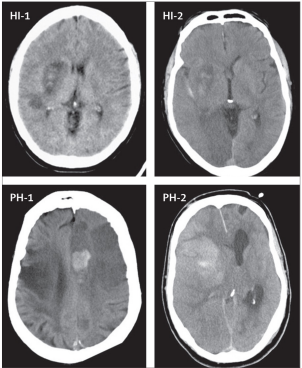
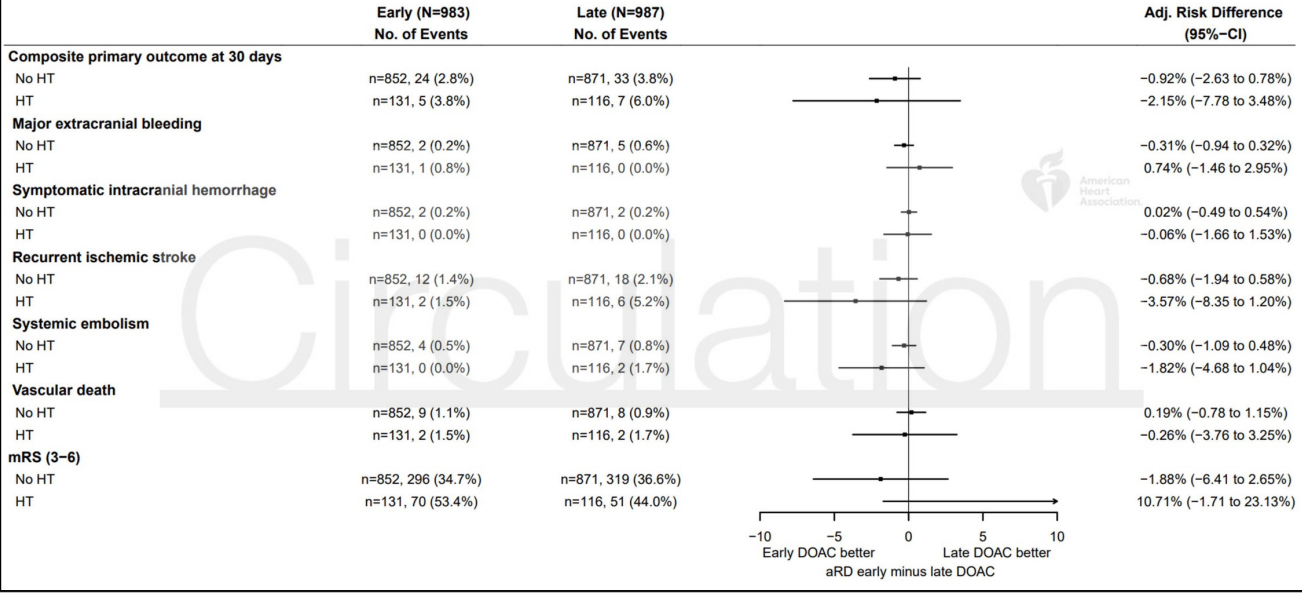
No treatment effect heterogeneity or elevation in sx ICH risk with early DOAC in subgroups: stroke severity, HT 1/2

Figure 2. Unadjusted Odds Ratios (ORs) of the Primary Composite Outcome and Secondary Outcomes at 30 Days



Hemorrhagic infarction - 1	Isolated petechial staining of infarcted tissue without mass effect
Hemorrhagic infarction - 2	Onfluent patchy in infarcted tissue without mass effect
Parenchymal Hemorrhage - 1	Homogeneous high attenuation lesion with minimal mass effect occupying less than 30% of the infarcted area
Parenchymal Hemorrhage - 2	Lesion occupying more than 30% of the infarcted area with definite mass effect
	Possible extension of hemorrhage into the ventricular space, As well as any hemorrhage outside the infarcted area

Figure 3: Haemorrhagic infarction and parenchymal haemorrhage



Early DOAC may worsen functional outcomes in IS with PH

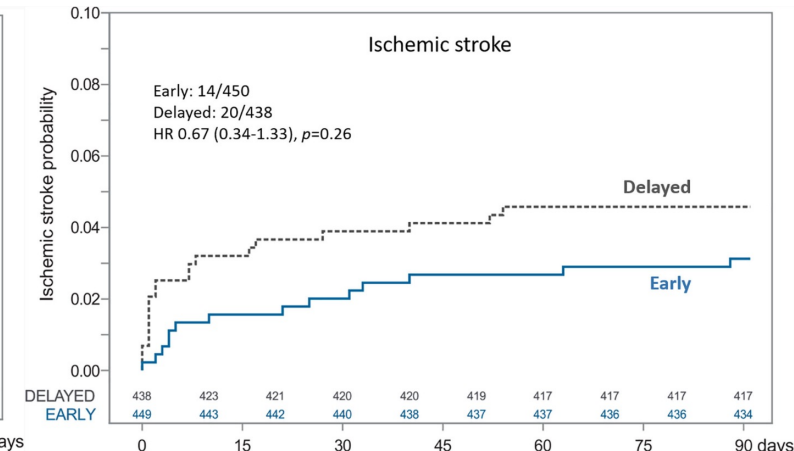
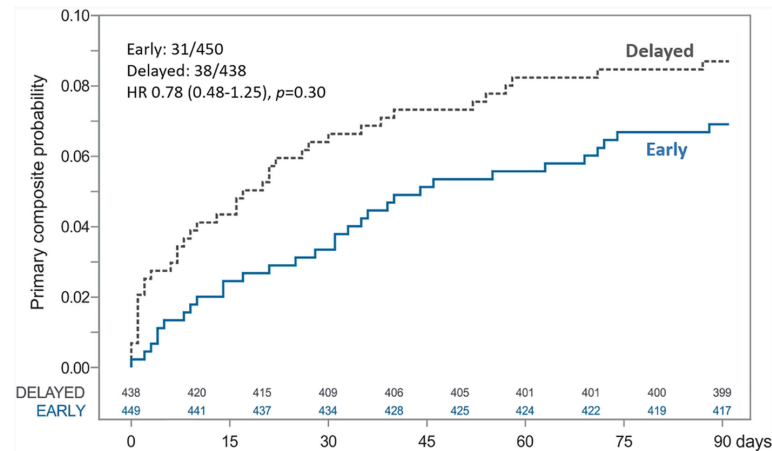
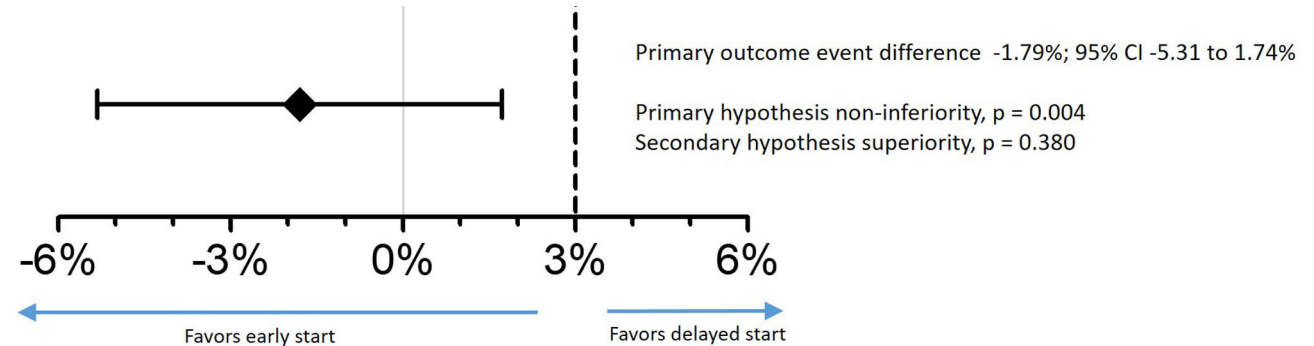
Timing of OAC Therapy in Acute IS w/ AF (TIMING)

- *Population*: acute ischemic stroke + AF
- *Intervention/Control*: early/late DOAC
 - Early: within 4 days (not 72 hours).
 - Delayed: 5-10 days
- *1° Outcome*: composite of recurrent IS, sx ICH, death @ 90d

Noninferiority margin: 3%; met.

Superiority was not demonstrated.

- Terminated early
- NIHSS: median 4 (IQR 2-9)



IS despite AC

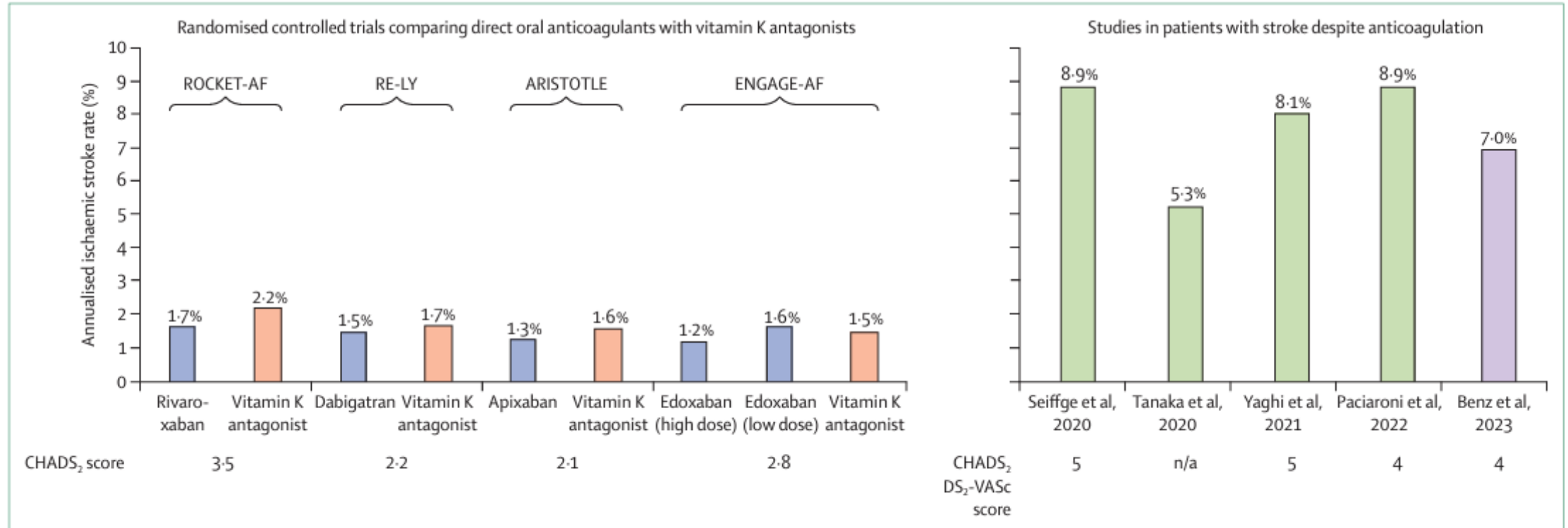


Figure: Annualised rates of ischaemic stroke events in randomised controlled trials comparing vitamin K antagonists (red bars) with different direct oral anticoagulants (blue bars) and in studies of patients with stroke despite anticoagulant therapy (green bars [observational studies] and violet [IPDMA of the RCTs RE-LY, ARISTOTLE, ROCKET-AF, ENGAGE-AF, and AVERROES])

- Non-adherence
- Alternative high-risk mechanism
- Inappropriate dosing
- Drug-drug interactions
 - CYP3A4 / P-glycoprotein efflux pump inducers: ↓ DOAC effect
 - Carbamazepine, phenytoin, primidone, phenobarbital
 - Rifampin
 - Drug-food interaction (rivaroxaban 15mg/20mg: bioavailability ↑ w/ food)

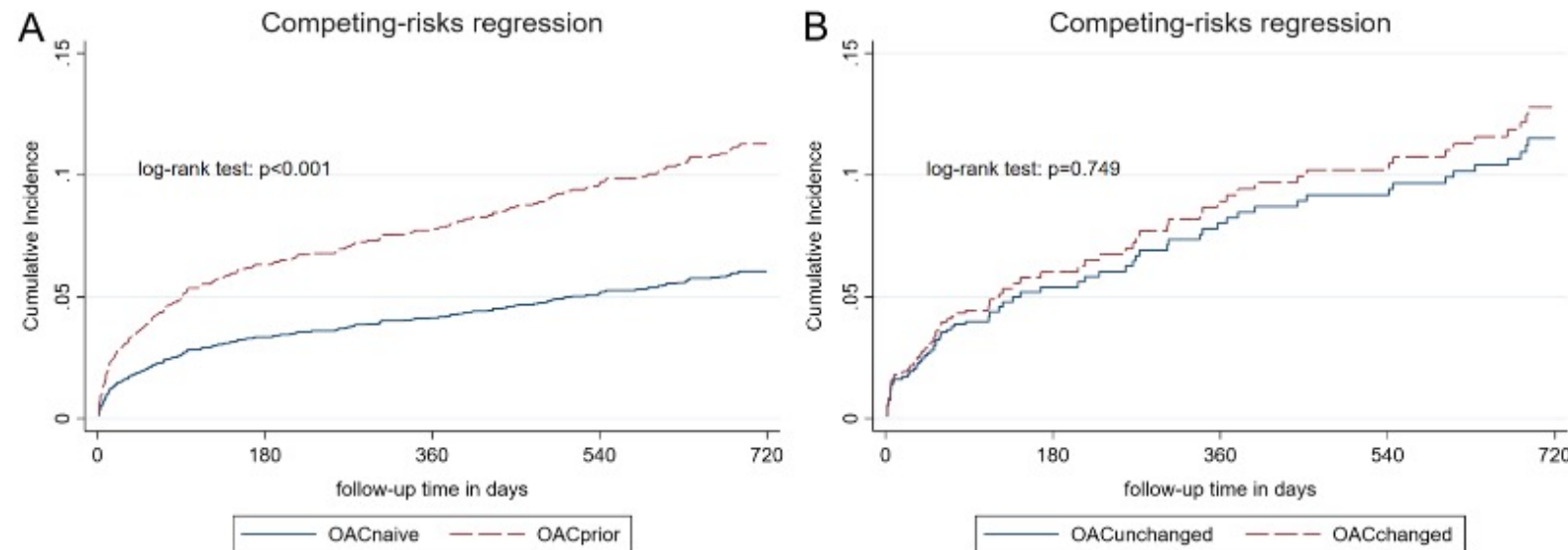


FIGURE 2: Cumulative incidence function curves for the main outcome of recurrent acute ischemic stroke. (A) Primary analysis of patients taking oral anticoagulation prior to the index event (OAC_{prior}, dashed line) compared to those not taking anticoagulants prior to the index event (OAC_{naive}, solid line). (B) Secondary analysis of patients that changed the type of anticoagulation (OAC_{changed}, dashed line) compared to those who continued the same type of anticoagulation (OAC_{unchanged}, solid line). [Color figure can be viewed at www.annalsofneurology.org]

B. Association of stroke etiology with the primary and secondary endpoint												
Stroke etiology	Composite outcome						Recurrent ischemic stroke					
	unadjusted			adjusted†			unadjusted			adjusted†		
	OR [95%-CI]	p value	N events/ total N in model	aOR [95%-CI]	p value	N events/ total N in model	OR [95%-CI]	p value	N events/ total N in model	aOR [95%-CI]	p value	N events/ total N in model
competing stroke mechanism	0.90 (0.69 to 1.16)	0.4	516/1906	1.18 (0.83 to 1.66)	0.363	473/1773	1.80 (1.07 to 3.02)	0.026	84/1842	1.83 (1.05 to 3.20)	0.034	77/1697
insufficient anticoagulation	1.02 (0.81 to 1.28)	0.891		0.93 (0.68 to 1.27)	0.648		0.91 (0.52 to 1.60)	0.751		0.99 (0.55 to 1.79)	0.968	
cardioembolism despite sufficient anticoagulation	(reference)			(reference)			(reference)			(reference)		

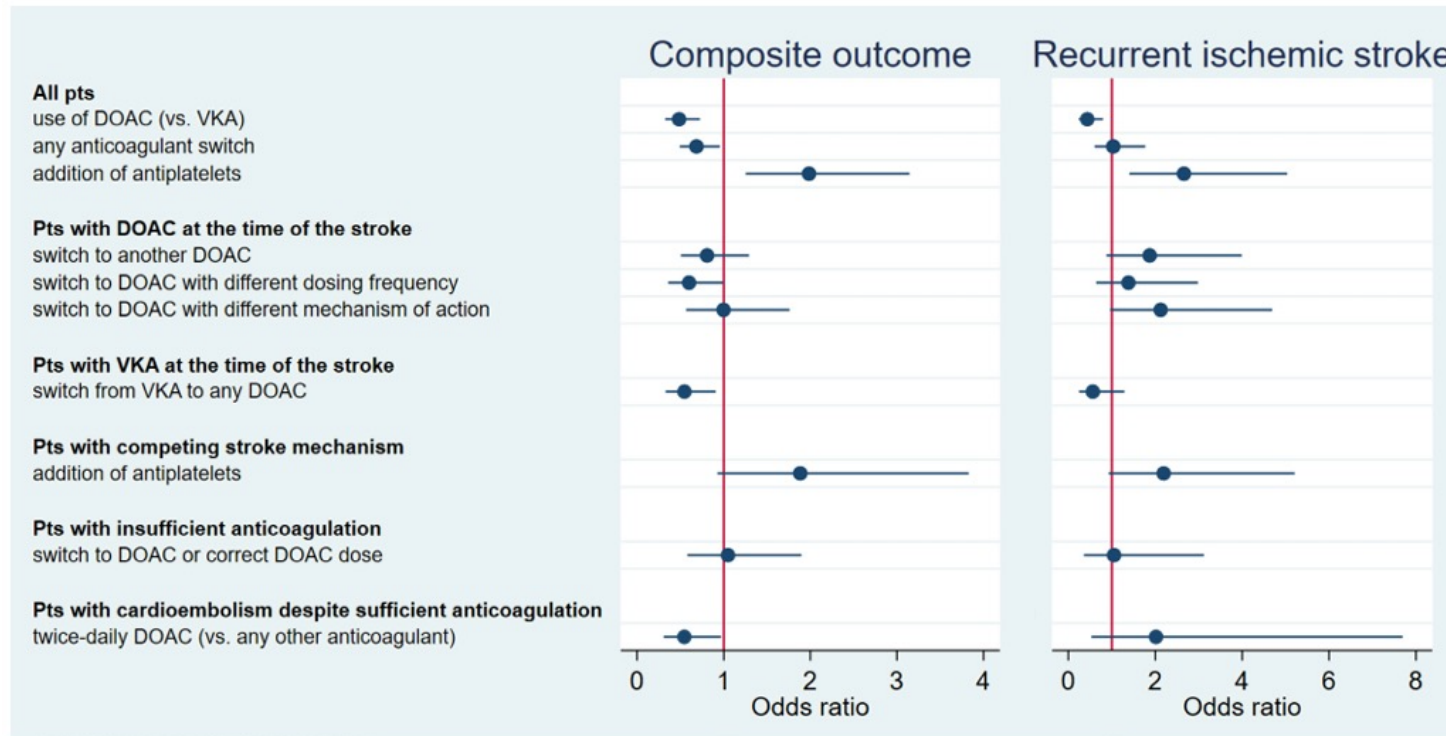
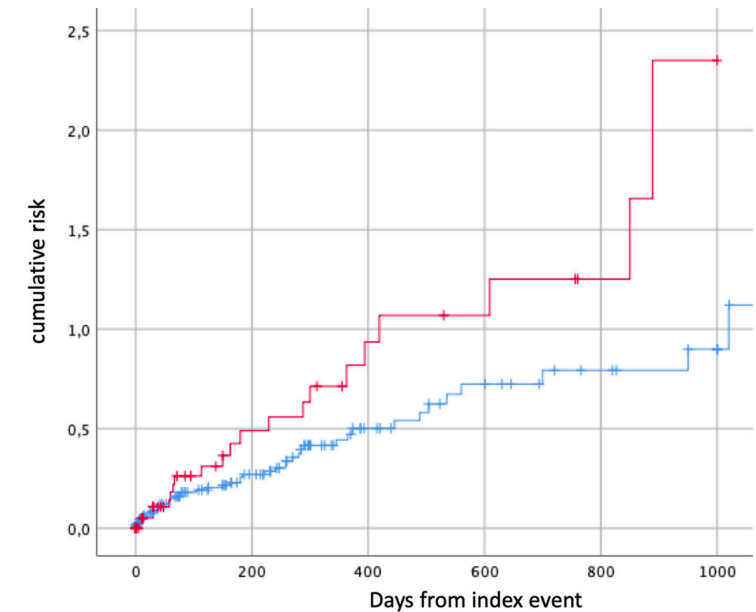


Figure 3 Association of preventive strategies after stroke despite anticoagulation with the primary and secondary endpoints from the adjusted models.



Hemorrhagic endpoint: intracranial bleeding and major extracranial bleeding (NOAC with or without antiplatelet therapy)
OR 2.8 (95% 1.4-5.5)
Log Rank (Mantel – Cox): p=0.04

Part 1 - Key Messages

- Ensure comorbidities & symptom mgmt. addressed
- DOACs (apixaban) preferred
 - Warfarin exceptions: mechanical valves; rheumatic mitral stenosis or mitral stenosis of moderate or greater severity.
- Concurrent AP tx indicated → DOAC + P2Y12 inhibitor
- Assess (and re-assess) bleeding risk
- Early DOAC for mild-moderate infarcts appears safe & efficacious
- Switching 1 DOAC to another has unclear benefit when IS occurs despite AC

Novel Approaches

- Left atrial appendage occlusion
- Permanent CCA filters
- Early rhythm control
- Novel OACs

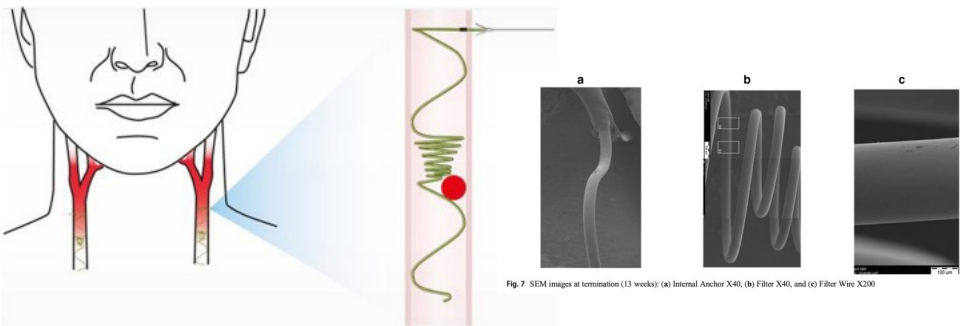
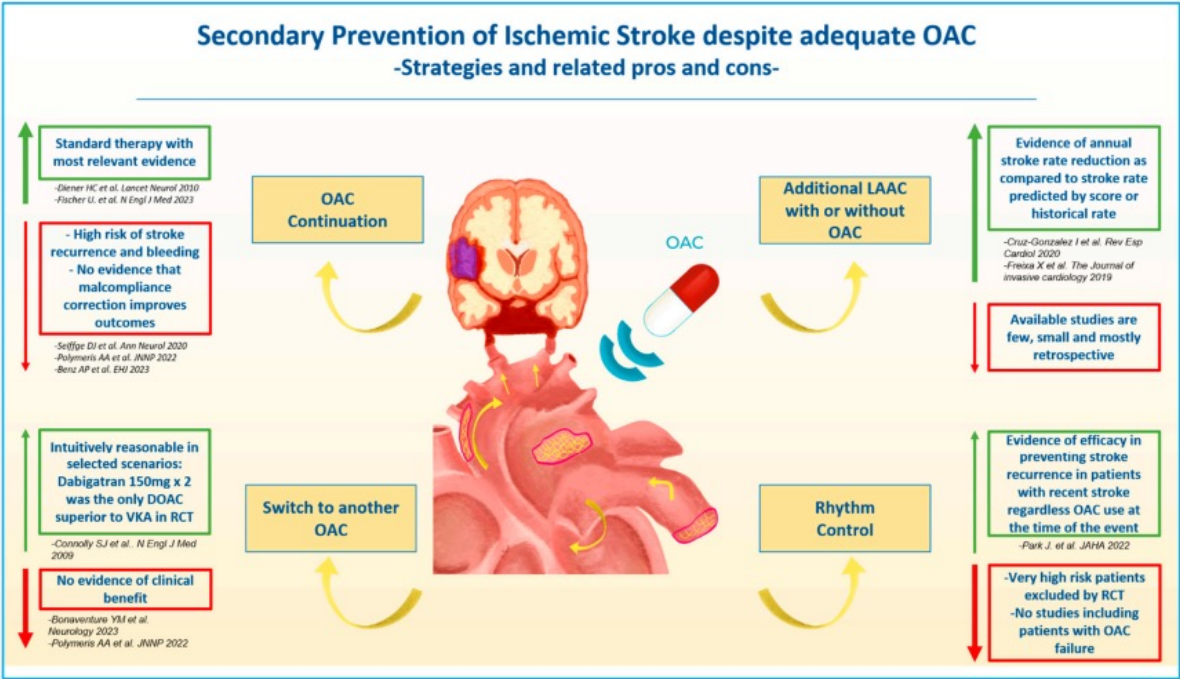


Fig. 7 SEM images at termination (11 weeks): (a) Internal Anchor X40, (b) Filter X40, and (c) Filter Wiro X200

J Clin Med. 2023 Sep 5;12(18):5784.
 Lancet Neurol. 2024 Apr;23(4):404-417.
 Curr Cardiol Rep. 2020 Sep 10;22(11):144.



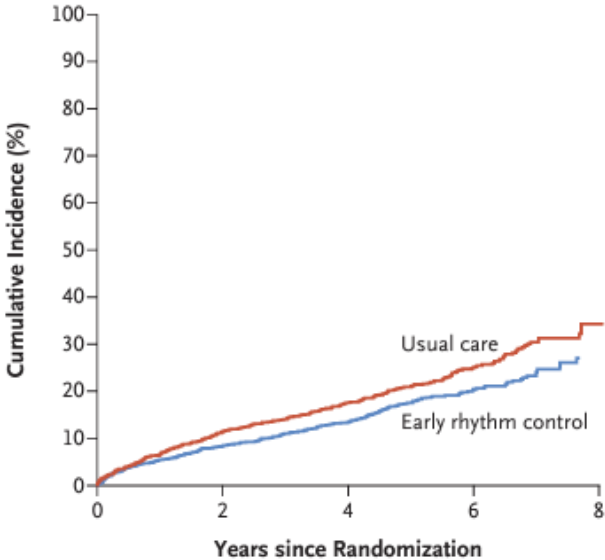
	INTERCEPT (NCT05723926)	ELAPSE (NCT05976685)	LAAOS-4 (NCT05963698)
Intervention	Permanent bilateral carotid artery filter on top of DOAC therapy (plus antiplatelet therapy for 6 months)	Percutaneous left atrial appendage occlusion (any approved device) on top of DOAC therapy (plus antiplatelet therapy for 6 weeks)	Percutaneous left atrial appendage occlusion with device
Comparator	Standard DOAC therapy	Standard DOAC therapy	Local, standard medical care (any anticoagulation)
Sample size	2000 participants	482 participants (adaptive design)	4000 participants
Estimated percentage of patients with breakthrough strokes	66%	100%	Unknown (patients with breakthrough stroke are eligible among other patients at high risk of stroke)
Primary endpoint	Large vessel anterior circulation ischemic stroke	Composite of ischaemic stroke, systemic embolism, and cardiovascular death	Composite of ischaemic stroke and systemic embolism
Follow-up	44 months	Minimum 6 months, and maximum 48 months	4 years (estimated average)
Current status	Vanguard trial in preparation	First patient recruited in Q1/2024	First patient recruited in Q4/2023

DOAC=direct oral anticoagulant therapy.

Table 3: Trials in patients with atrial fibrillation and stroke despite anticoagulation therapy

Rhythm Control

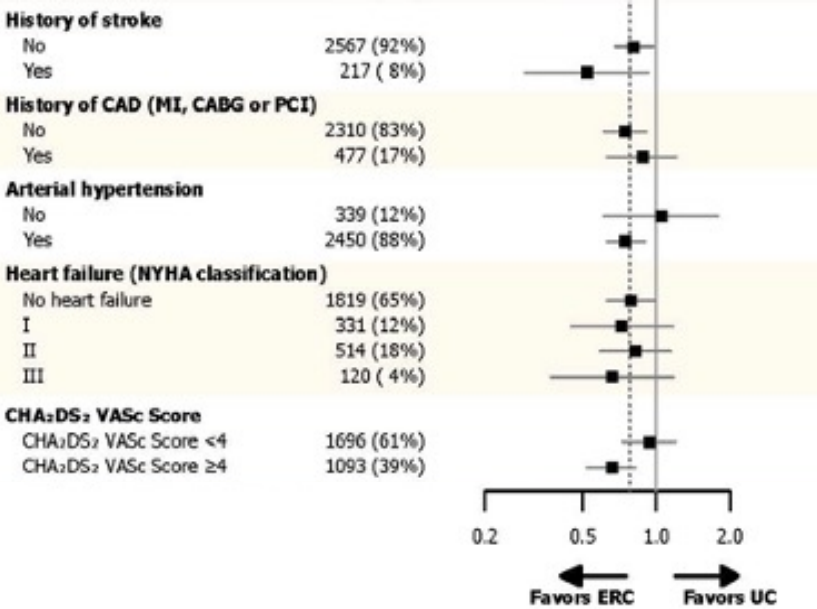
	Rhythm control
	EAST-AFNET 4 (NCT01288352) ^{66,67}
Trial characteristics	
Sample size	2789 participants
Intervention	Antiarrhythmic drugs or ablation
Comparator	No rhythm control
Anticoagulation	2517 (90%) participants
History of ischaemic stroke	328 (12%) participants
Outcomes	
Follow-up	5.1 years
Primary endpoint	Primary death from cardiovascular causes, stroke, or hospitalisation with worsening of heart failure or acute coronary syndrome 3.9 vs 5.0 events per 100 patient-years HR 0.79 (96% CI 0.66–0.94; p=0.005)
Stroke	0.6% (rhythm control) vs 0.9% (no rhythm control)
Mortality	1.0% (rhythm control) vs 1.3% (no rhythm control)



No. at Risk					
Usual care	1394	1169	888	405	34
Early rhythm control	1395	1193	913	404	26

Figure 2. Aalen–Johansen Cumulative-Incidence Curves for the First Primary Outcome.

Table 2. Efficacy Outcomes.*			
Outcome	Early Rhythm Control	Usual Care	Treatment Effect
First primary outcome — events/person-yr (incidence/100 person-yr)	249/6399 (3.9)	316/6332 (5.0)	0.79 (0.66 to 0.94)†
Components of first primary outcome — events/person-yr (incidence/100 person-yr)			
Death from cardiovascular causes	67/6915 (1.0)	94/6988 (1.3)	0.72 (0.52 to 0.98)‡
Stroke	40/6813 (0.6)	62/6856 (0.9)	0.65 (0.44 to 0.97)‡
Hospitalization with worsening of heart failure	139/6620 (2.1)	169/6558 (2.6)	0.81 (0.65 to 1.02)‡
Hospitalization with acute coronary syndrome	53/6762 (0.8)	65/6816 (1.0)	0.83 (0.58 to 1.19)‡



RAFAS (NCT02285387) ¹⁶	
Trial characteristics	
Sample size	273 participants
Intervention	Antiarrhythmic drugs or ablation
Comparator	No rhythm control
Anticoagulation	255 (94%) participants
History of ischaemic stroke	273 (100%) participants
Outcomes	
Follow-up	Outcome assessement for all participants at 12 months
Primary endpoint	Recurrent stroke at 3 months (2 [1.1%] vs 4 [4.2%]; HR, 0.257 [log-rank p=0.091]) and at 12 months (3 [1.7%] vs 6 [6.3%]; HR 0.251 [log-rank p=0.034])
Stroke	See primary endpoint
Mortality	Not reported

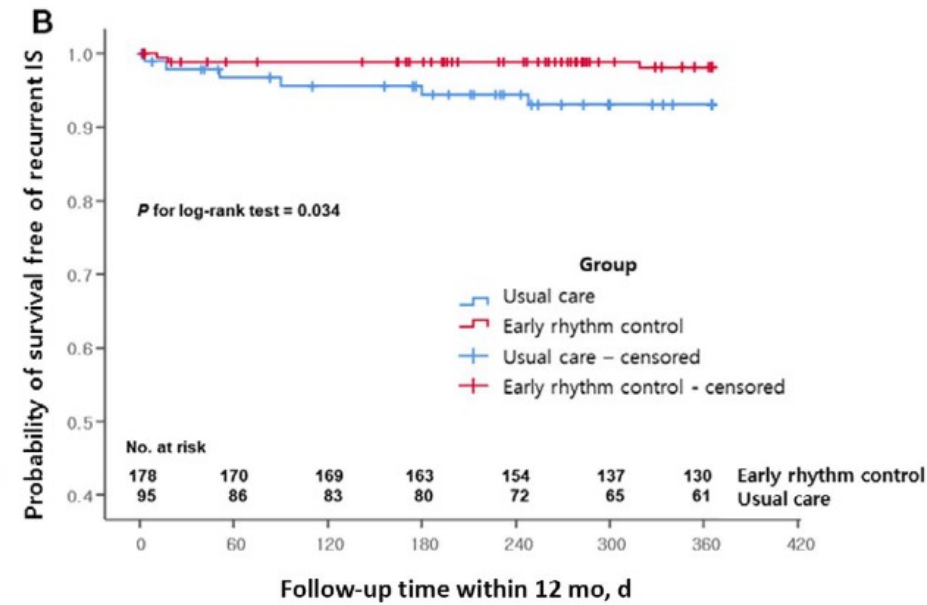
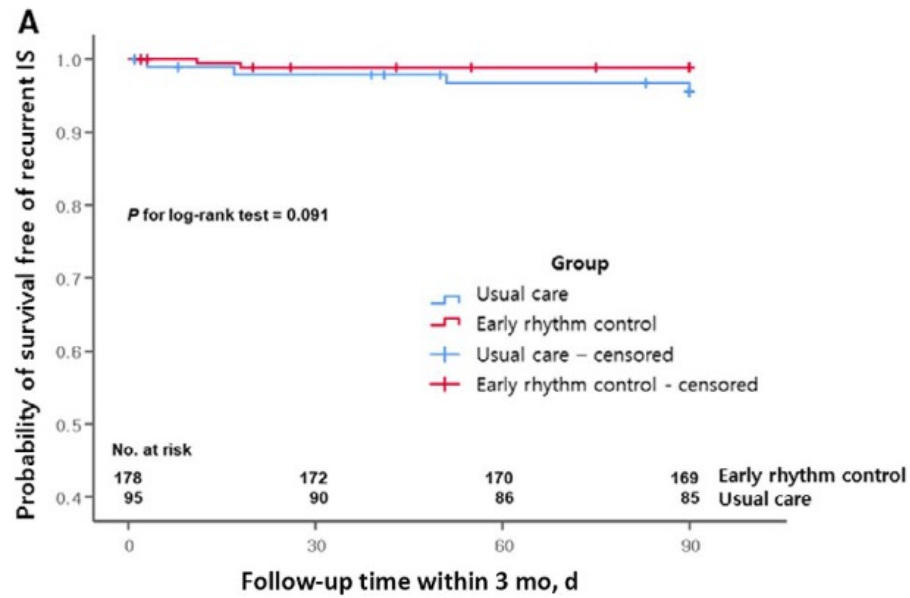


Table 2. Primary and Secondary Clinical Outcomes

	Early rhythm control (n=178)	Usual care (n=95)	HR	95% CI	<i>P</i> value
Primary outcome, n (%) [*]					
Recurrent stroke in 3 mo	2 (1.1)	4 (4.2)	0.257	0.047–1.405	0.117
Recurrent stroke in 12 mo	3 (1.7)	6 (6.3)	0.251	0.063–1.003	0.050
Secondary outcome, n (%) ^{**}					
Composite outcome in 3 mo	19 (10.7)	10 (10.5)	0.995	0.463–2.140	0.990
Composite outcome in 12 mo	25 (14.0)	16 (16.8)	0.808	0.431–1.513	0.505
Arrhythmia-related events in 3 mo	3 (1.7)	0	NA	NA	NA
Arrhythmia-related events in 12 mo	5 (2.8)	1 (1.1)	2.565	0.3–21.958	0.390
Sustained AF	60 (34.1)	59 (62.8)			<0.001
AF detection period in the consecutive Holter during 12 mo, mo	3.0 (1.0–9.0)	7.0 (1.0–12.0)			0.002
Stroke to NSR duration, d	13.0 (2.0–84.0)	2.0 (0.0–98.5)			0.083

Direct Factor XI & XIa Inhibitors

- Factor XI → differential role in hemostasis & thrombosis

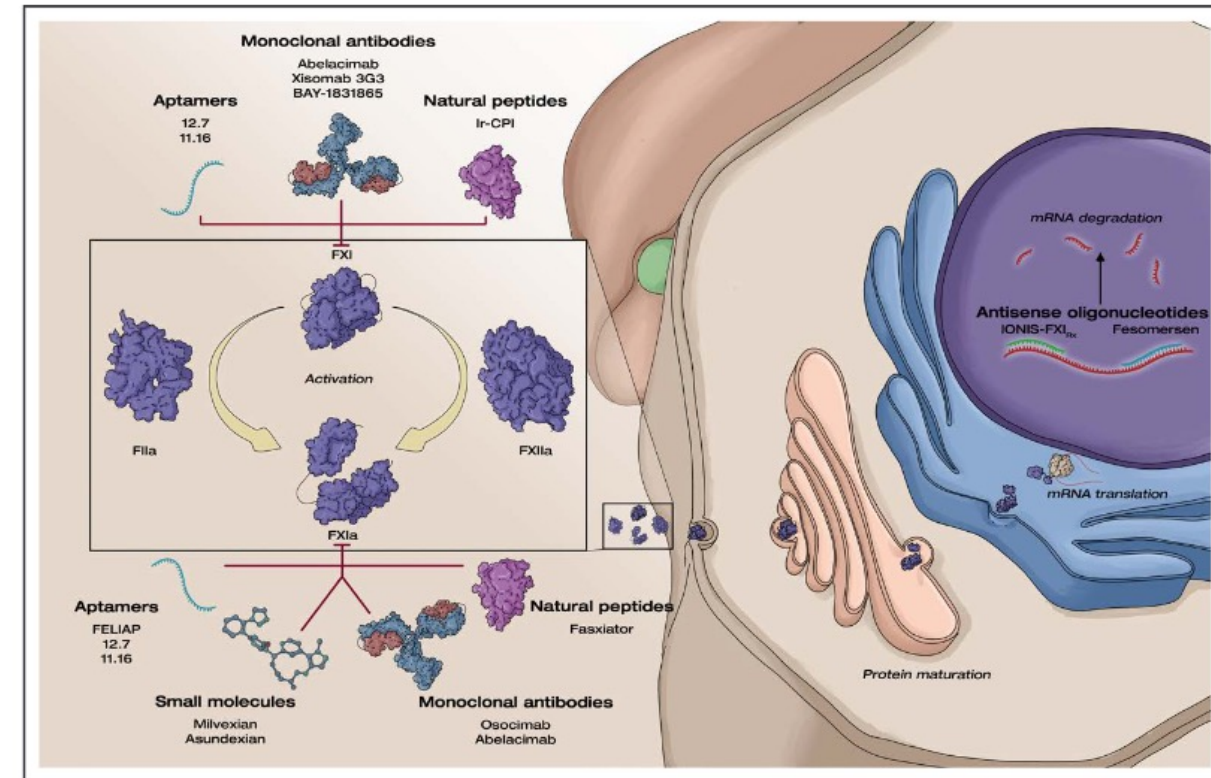
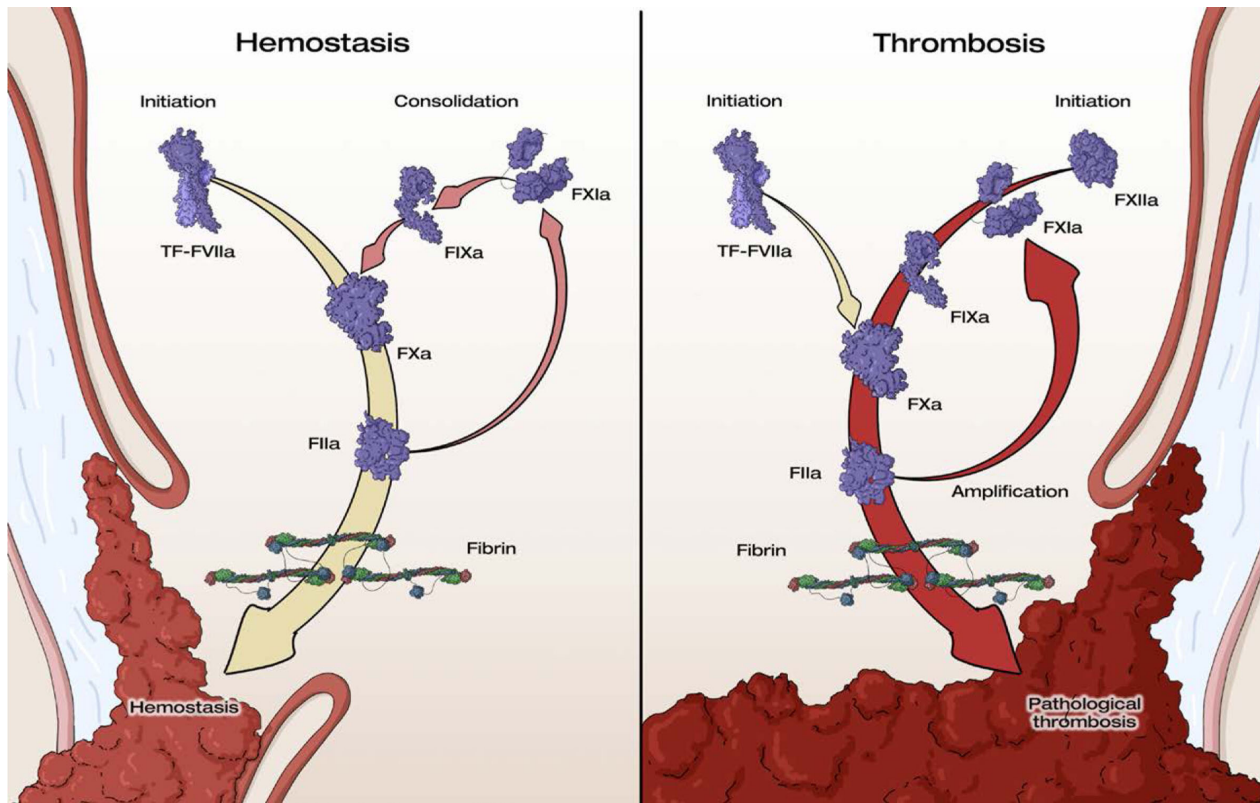


Figure 2. Classification and mechanism of action of FXI inhibitors.

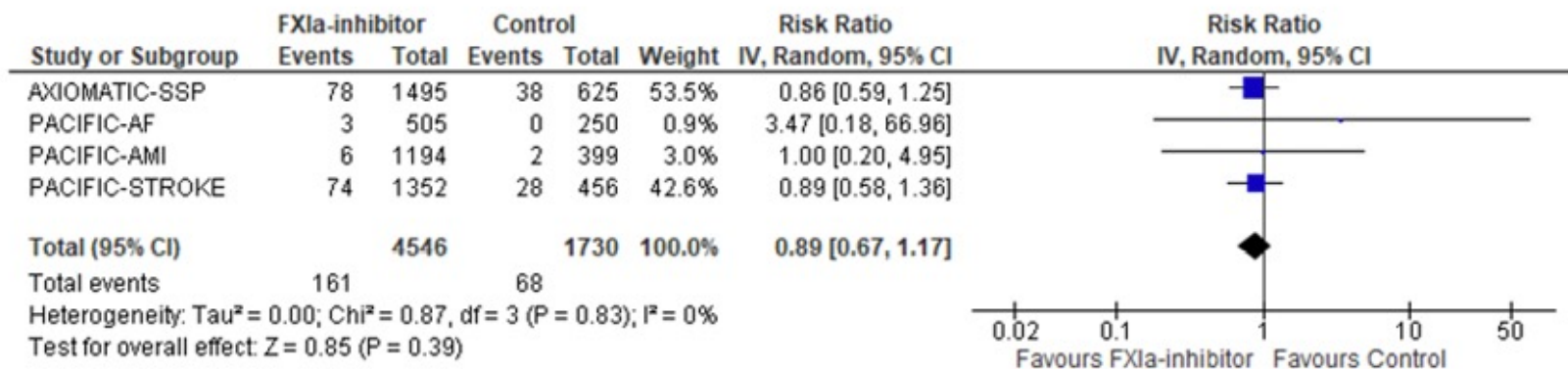


Figure 3. Forest plot presenting the association of factor XIa inhibitors compared to controls with symptomatic ischemic stroke.

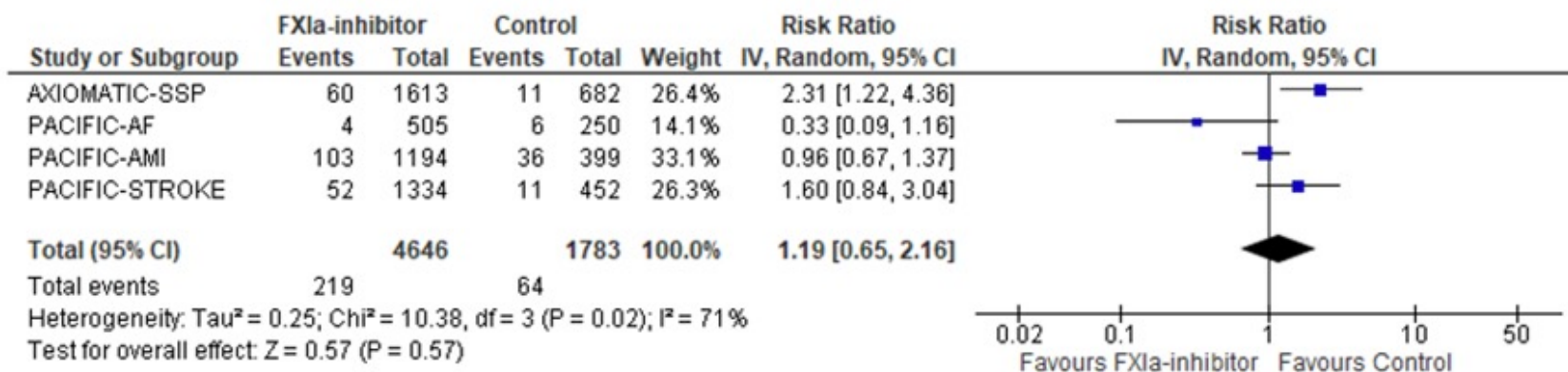


Figure 4. Forest plot presenting the association of factor XIa inhibitors compared to controls with the composite of major bleeding and clinically relevant non-major bleeding.

- Phase 3 AF trials
 - OCEANIC-AF (asundexian) terminated → lack of efficacy
 - LIBREXIA-AF (milvexian) → active
 - LILAC-TIMI 76 (abelacimab) → active
 - Phase 2b: 67% ↓ risk of hemorrhage vs. rivaroxaban (N=1287)
- AZALEA: prior ischemic stroke in 191/1,287 (14.9%).

Cardiac Rhythm Evaluation – AF after IS

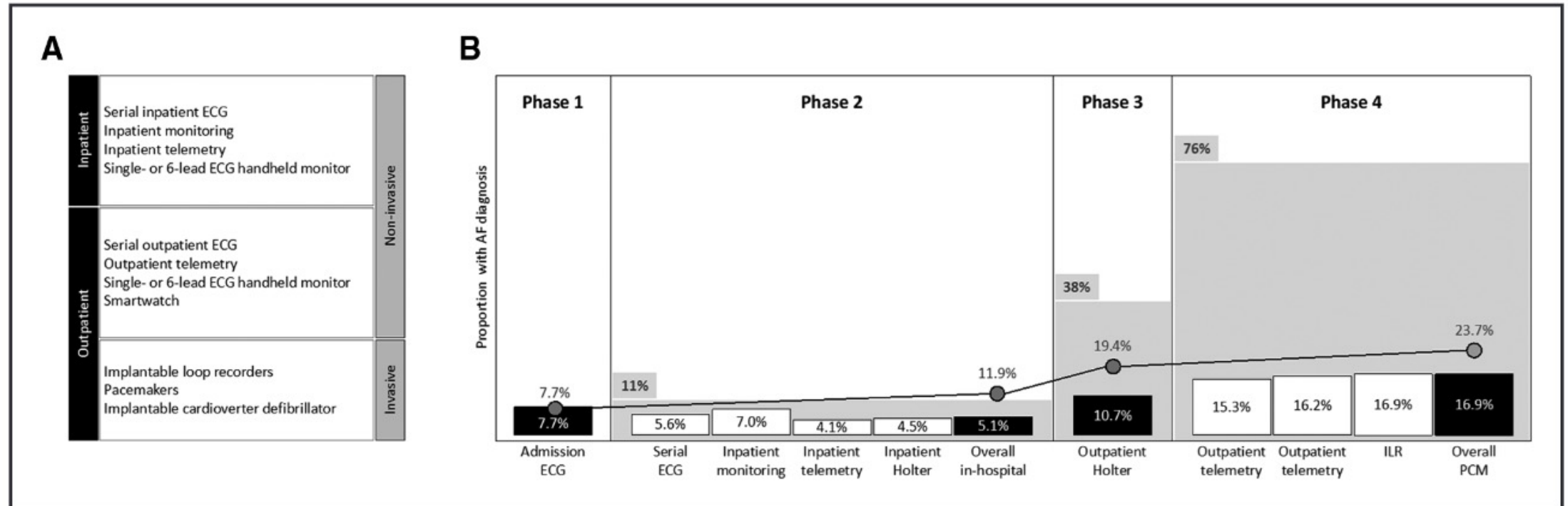


Figure. Available cardiac rhythm monitoring technologies and their incremental diagnostic yield.

- Range 24 hours – 3 years
- RCTs → designed/powered to detect Δ AF rates (not IS rates)

Table. Rate of AF Detection in Clinical Trials of Prolonged (14 Days or More) Cardiac Monitoring in Patients With Recent Ischemic Stroke

Trial	Intervention	Duration of long-term monitor (d)	Required AF duration	AF rate (%)	Ischemic events (no.)
CRYSTAL-AF	ICM	180	>30 s	8.9	11
	24-h holter or telemetry	--	>30 s	1.4	18
EMBRACE	30-d event recorder	30	>30 s	16.1	NR
	24-h holter	--	>30 s	3.2	NR
Find-AF _{RANDOMIZED}	10-d holter repeated thrice	30	>30 s	14	5
	24-h holter	--	>30 s	5	9
PER-DIEM	ICM	180	>120 s	15.3	5
	External loop recorder	30	>120 s	4.7	8
STROKE-AF	ICM	365	>120 s	12.1	16
	Usual care	--	>30 s	1.8	23

Cryptogenic {

~50% Cryptogenic ←

66% Cryptogenic ←

cSVD or LAA stroke ←

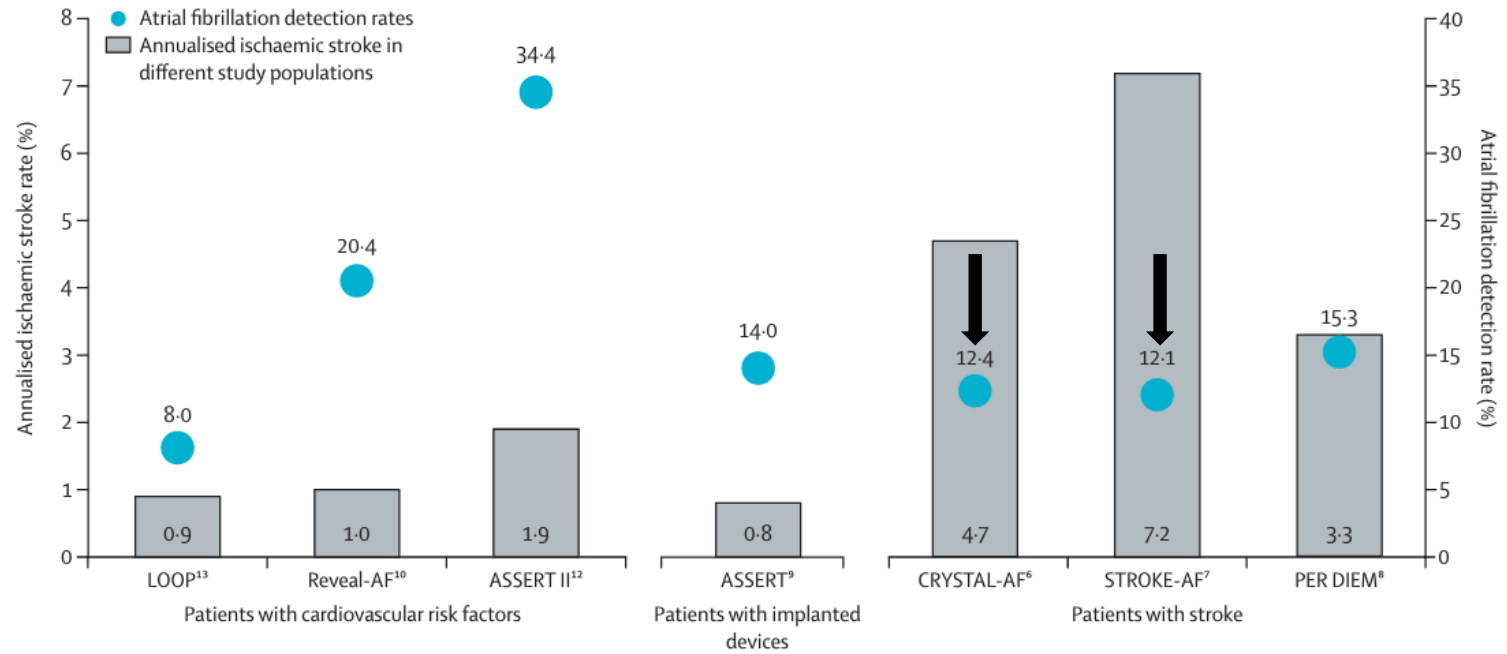


Figure 3 Forest Plot Presenting the Differences Between Prolonged and Conventional (Nonprolonged) Cardiac Rhythm Monitoring in the Probability of Anticoagulant Initiation, Stratified by Study Type

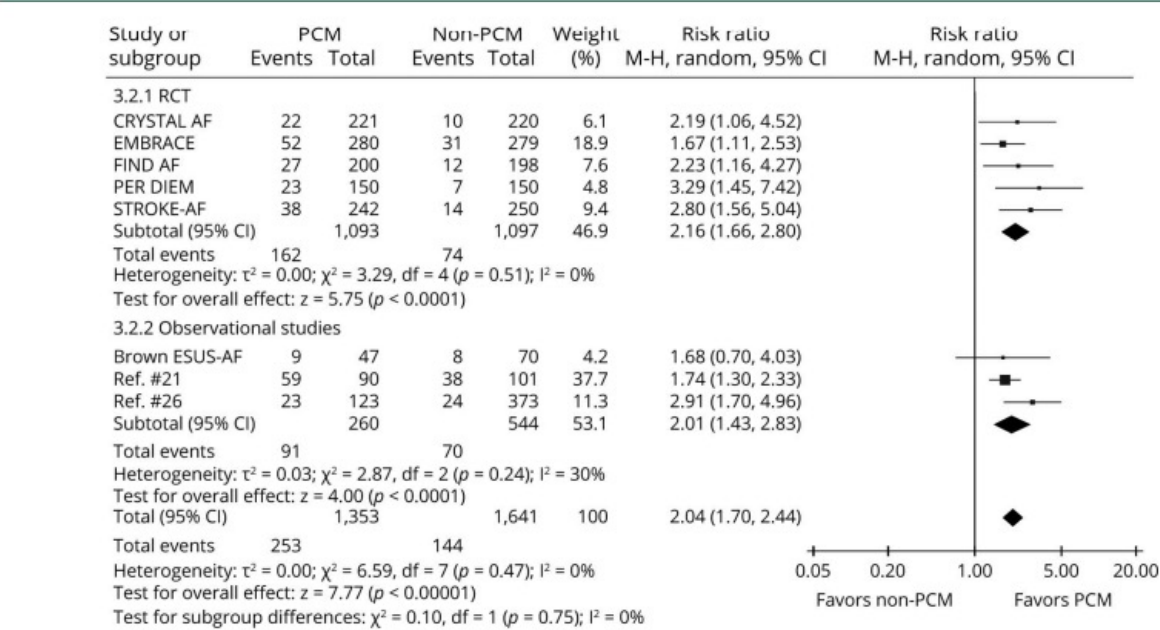
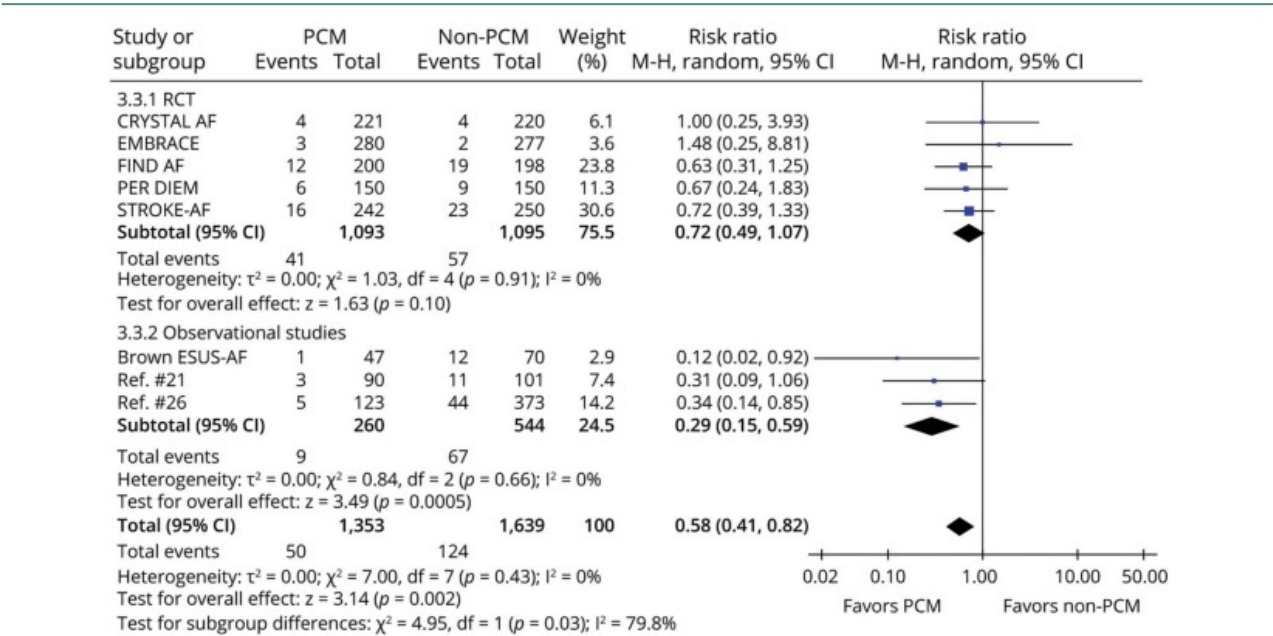


Figure 4 Forest Plot Presenting the Differences Between Prolonged and Conventional (Nonprolonged) Cardiac Rhythm Monitoring in the Risk of Recurrent Stroke During Follow-Up, Stratified by Study Type



Prolonged cardiac rhythm monitoring in stroke populations:

- Increased AC initiation
- Unclear effect on recurrent stroke risk
- AF burden often low

LOOP Study

- *Population*: 70-90 y + HTN / DM / prior stroke / HF
- *Intervention/Control*: ILR / usual care
- *1° Outcome*: stroke / systemic embolism
- * CHA₂DS₂-VASc: median 4 (IQR 3-4)
- * Prior stroke/TIA/SE: ~25%
- * AF detection: ILR ~32%, usual care ~12%

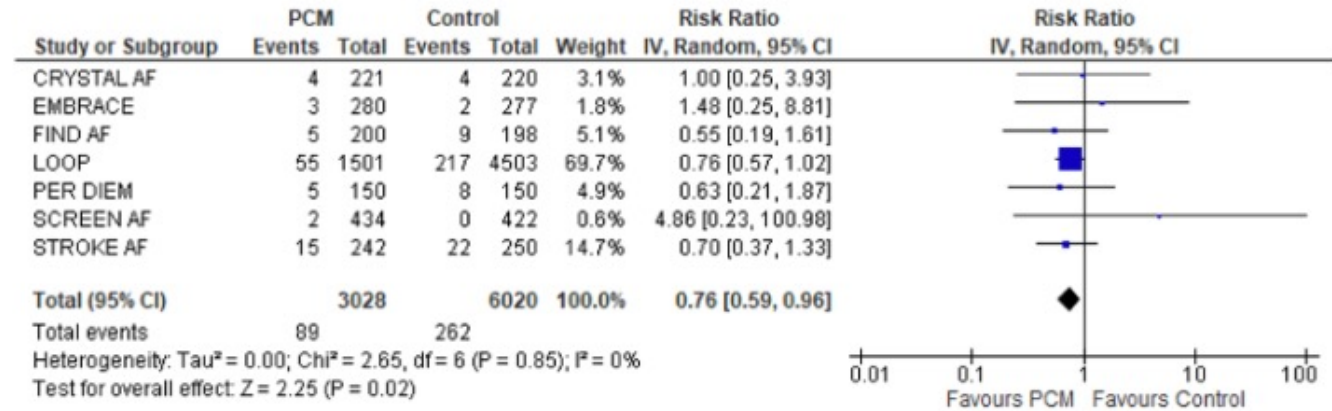


Figure 2. Forest plot presenting the association of prolonged cardiac monitoring (PCM) with occurrence of ischemic stroke compared to controls.

A Stroke or systemic arterial embolism

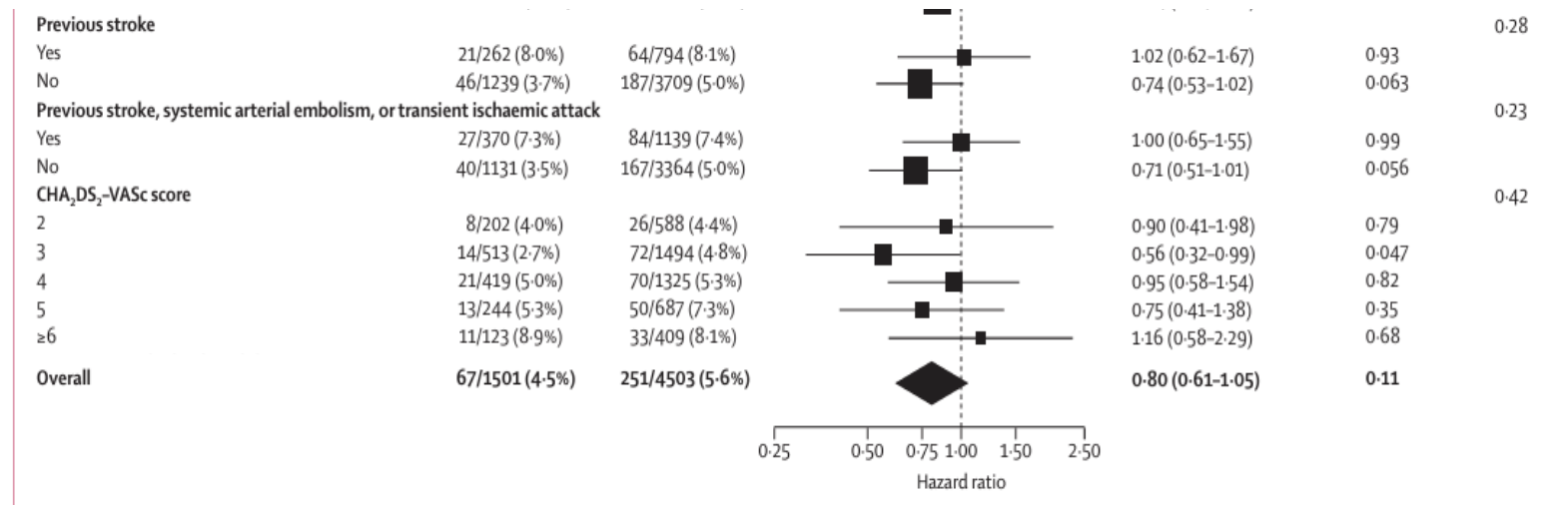
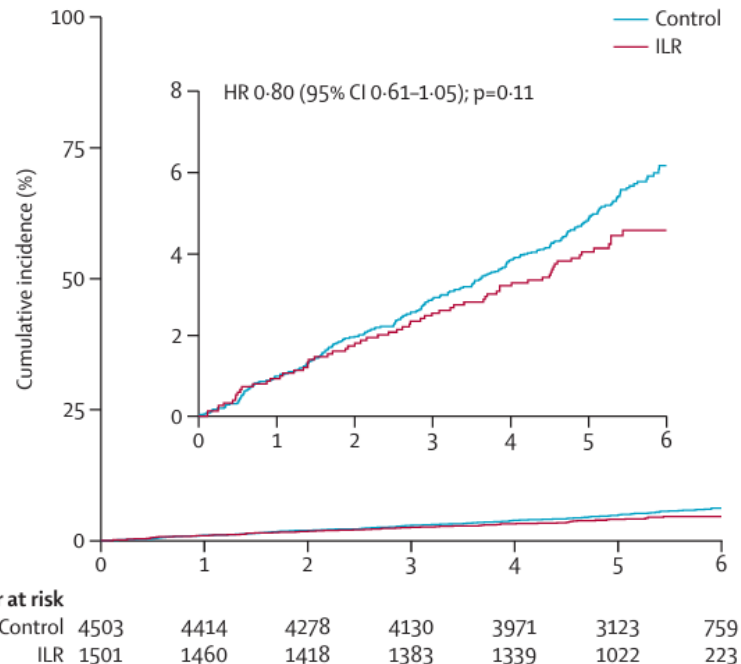
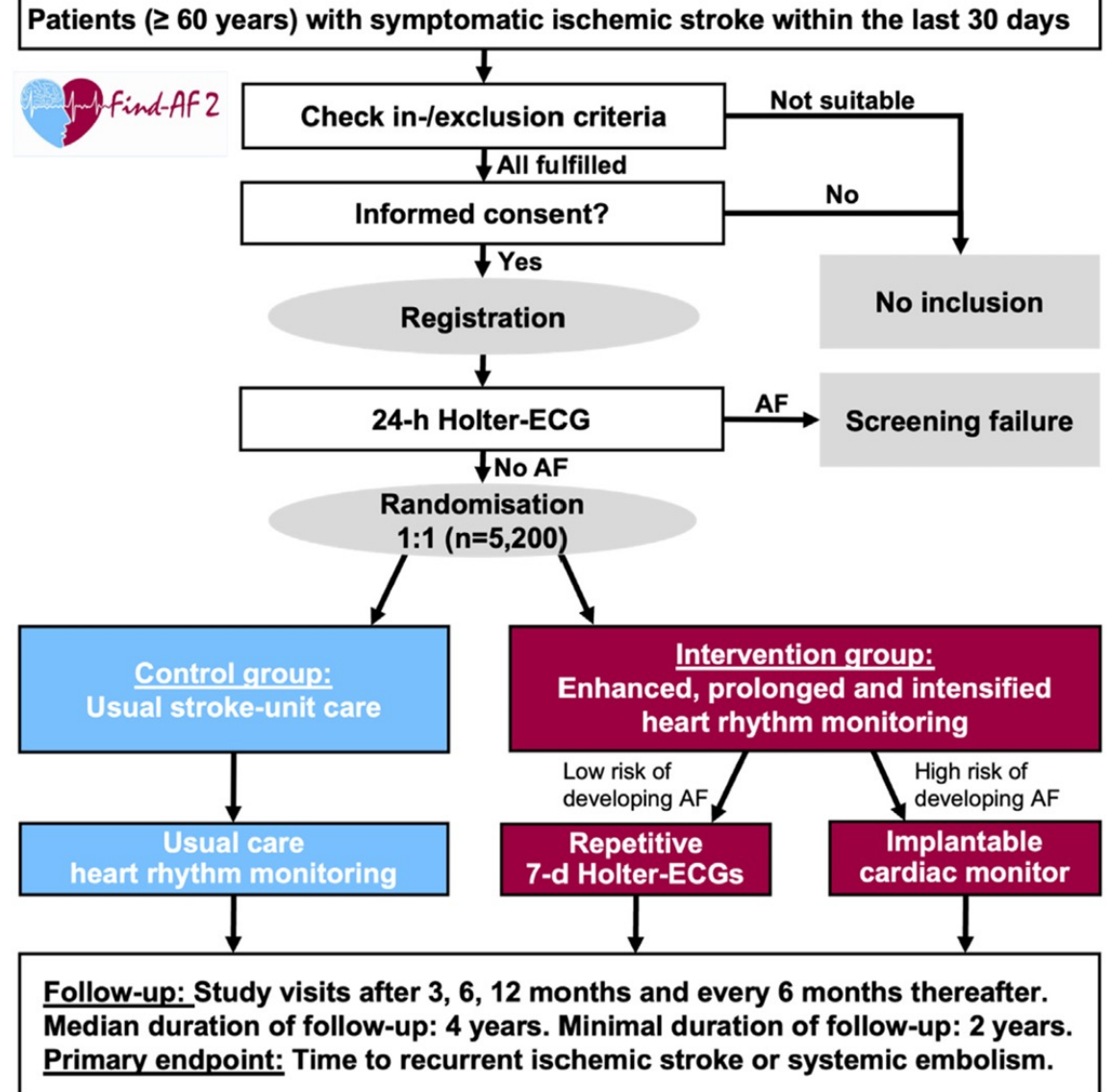
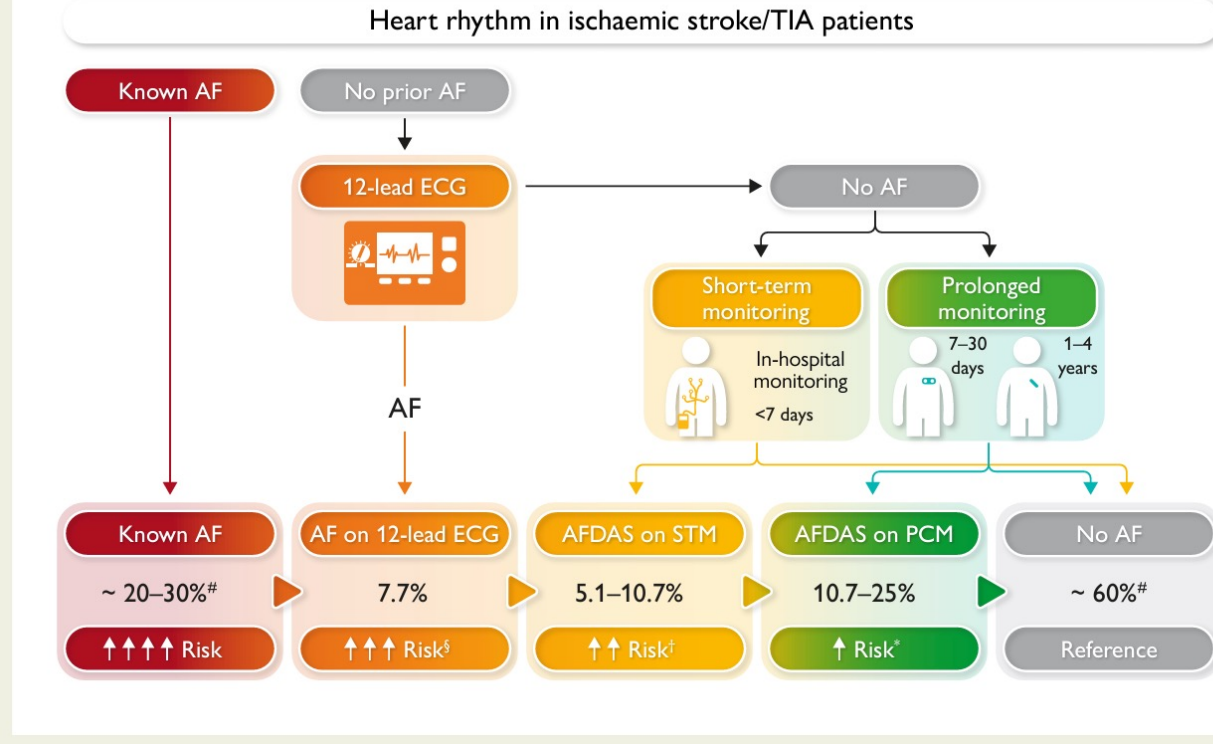


Figure 3: Frequency of the primary outcome grouped by randomisation arm and hazard ratios in prespecified subgroups

Lancet. 2021 Oct 23;398(10310):1507-1516.

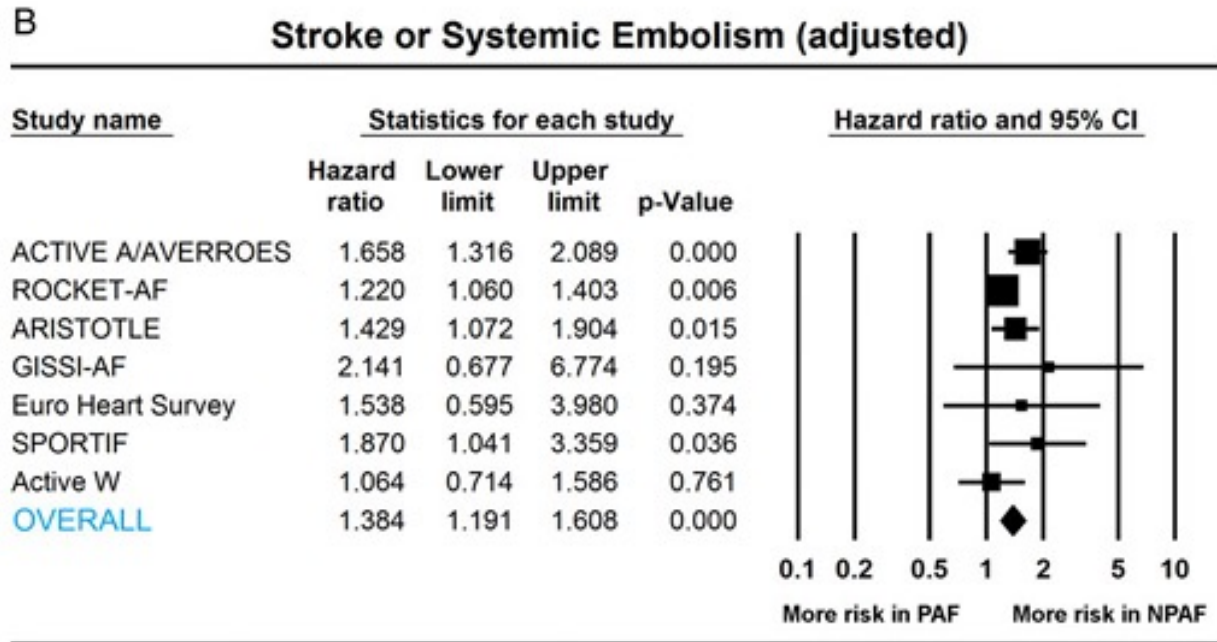
Eur Stroke J. 2023 Mar;8(1):106-116.

Graphical Abstract



AF Burden & IS Risk

- IS risk ↑ permanent > paroxysmal AF



Eur Heart J. 2016 May 21;37(20):1591-602.
 Eur Heart J. 2017 May 1;38(17):1339-1344.
 Eur Heart J. 2014 Feb;35(8):508-16.
 Circulation. 2016 Oct 18;134(16):1130-1140.
 Circulation. 2019 Nov 12;140(20):1639-1646.

- Predictors of AF burden
 - Age
 - Left atrial enlargement
 - Natriuretic peptides
 - Frequent PACs
- Burden of paroxysmal AF ~ IS risk
 - ASSERT (post-hoc analysis) → 24h
 - SOS-AF project → 1h
- CHA₂DS₂-VASc X AF Burden interaction

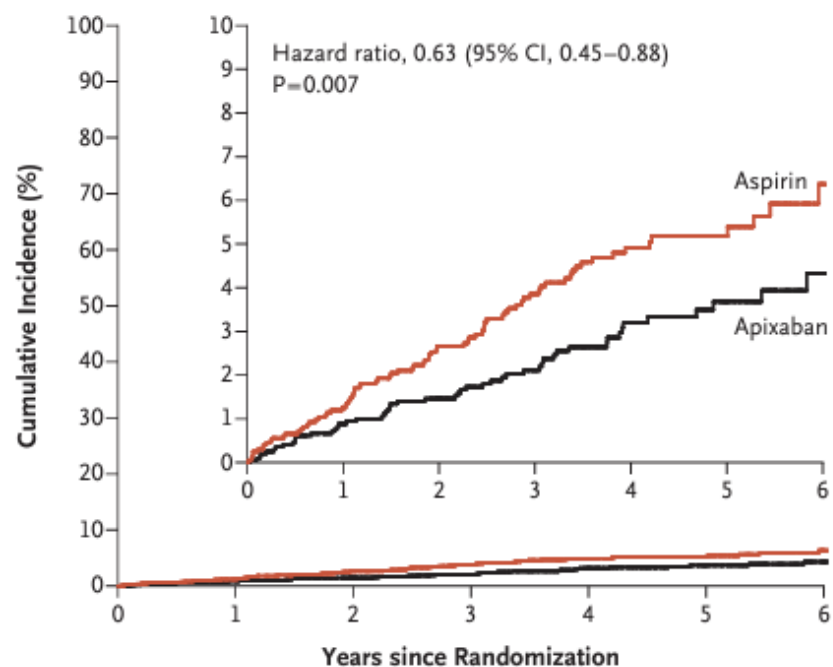
RATE: brief episodes were not associated with clinical events.*

*Not a validated treatment threshold; full duration definition unverified. Repair 16.

		CHA ₂ DS ₂ -VASc Score				
		0	1	2	3-4	≥5
		n=2922 (13.4%)	n=2151 (9.9%)	n=4554 (20.9%)	n=7164 (32.9%)	n=4977 (22.9%)
Maximum Daily AF Duration	No AF	0.33% 40 events	0.62% 46 events	0.70% 95 events	0.83% 139 events	1.79% 157 events
	n=16815 (77.2%)					
	AF 6 min–23.5 h	0.52% 11 events	0.32% 4 events	0.62% 17 events	1.28% 42 events	2.21% 36 events
	n=3381 (15.5%)					
AF >23.5h		0.86% 4 events	0.50% 3 events	1.52% 19 events	1.77% 28 events	1.68% 13 events
	n=1572 (7.2%)					

ARTESIA

- **Population:** $6m \leq AF \leq 24h$ detected by PM, defibrillator, cardiac monitor + $CHA_2DS_2-VASc \geq 3$, age $\geq 55y$
- **Intervention/Control:** apixaban / ASA
 - AF > 24h or clinical AF → study drug d/c'd & AC initiated
- **1° Outcome:** stroke / systemic embolism



No. at Risk							
Aspirin	1997	1777	1539	1120	780	468	200
Apixaban	2015	1786	1558	1157	820	474	214

Figure 1. Stroke or Systemic Embolism (Primary Efficacy Outcome).

Table 2. Clinical Outcomes (Intention-to-Treat Population).*

Outcome	Apixaban (N = 2015)		Aspirin (N = 1997)		Hazard Ratio (95% CI)	P Value
	no. of patients with event	%/patient-yr	no. of patients with event	%/patient-yr		
Stroke or systemic embolism	55	0.78	86	1.24	0.63 (0.45–0.88)	0.007
Stroke	55	0.78	84	1.21	0.64 (0.46–0.90)	
Ischemic or unknown type†	45	0.64	71	1.02	0.62 (0.43–0.91)	
Hemorrhagic	10	0.14	13	0.18	0.76 (0.33–1.73)	
Severity according to score on modified Rankin scale‡						
0–2	31	0.44	45	0.65	0.68 (0.43–1.07)	
3–6	19	0.27	37	0.53	0.51 (0.29–0.88)	
Missing data	5	0.07	2	0.03	2.48 (0.48–12.80)	
Systemic embolism	0		2	0.03	NA	
Stroke, TIA, or systemic embolism§	82	1.17	107	1.56	0.75 (0.56–1.00)	
Stroke, systemic embolism, or death from cardiovascular causes	148	2.10	171	2.47	0.85 (0.68–1.06)	
Stroke, myocardial infarction, systemic embolism, or death	419	6.01	418	6.10	0.98 (0.86–1.12)	
Myocardial infarction	37	0.52	41	0.59	0.89 (0.57–1.40)	
Death	362	5.06	341	4.82	1.04 (0.90–1.21)	
Death from cardiovascular causes	105	1.47	108	1.53	0.96 (0.73–1.25)	
Major bleeding¶	106	1.53	78	1.12	1.36 (1.01–1.82)	0.04
Fatal bleeding	10	0.14	14	0.20	0.70 (0.31–1.57)	
Symptomatic intracranial hemorrhage	17	0.24	23	0.33	0.73 (0.39–1.36)	
Gastrointestinal bleeding	55	0.78	31	0.44	1.76 (1.13–2.74)	
Transfusion performed	35	0.49	31	0.44	1.11 (0.68–1.80)	

* CHA_2DS_2-VASc : mean ~ 4 (≥ 4 : 61%)

* Prior stroke/TIA/SE: 9%

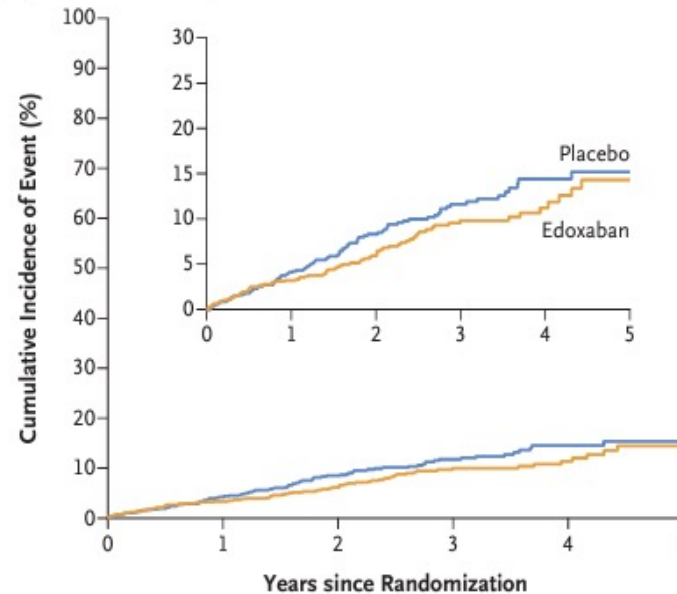
NOAH-AFNET 6

- Population: AHREs ≥ 6 m (implanted cardiac devices) + age ≥ 65 y + additional stroke risk factor
 - No AF
- Intervention/Control: edoxaban / placebo
- 1° Outcome: composite – stroke, SE, CV death
- * CHA₂DS₂-VASc: median 4 (IQR 3-5)
- Prior stroke/TIA: 10%
- AHREs duration: median 2.8h (IQR 0.8-9.4)
- ASA indication: ~54%

Table 2. Efficacy Outcomes.*

Outcome	Edoxaban (N = 1270)	Placebo (N = 1266)	Adjusted Hazard Ratio (95% CI)
	no. of patients with event/patient-yr (% per patient-yr)		
Primary composite efficacy outcome†	83/2557 (3.2)	101/2495 (4.0)	0.81 (0.60 to 1.08)‡
Ischemic stroke	22/2573 (0.9)	27/2519 (1.1)	0.79 (0.45 to 1.39)

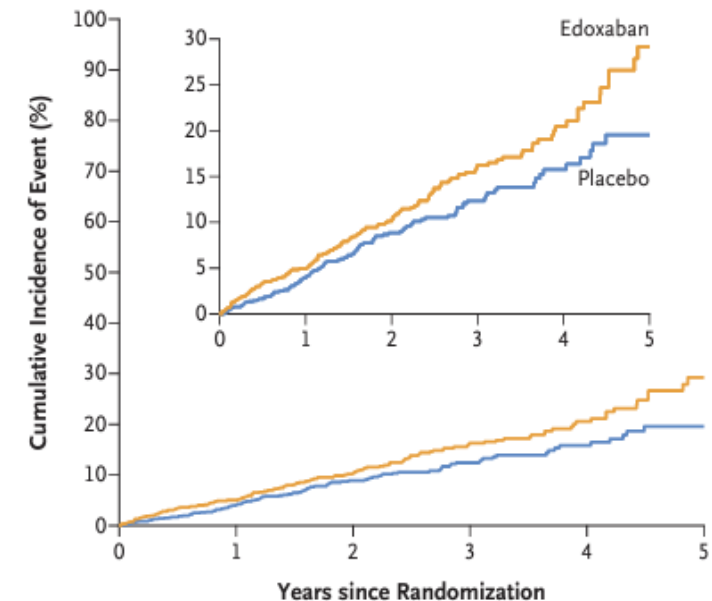
A Stroke, Systemic Embolism, or Death from Cardiovascular Causes



No. at Risk (no. of events)

Edoxaban	1270 (37)	873 (20)	559 (19)	327 (3)	148 (4)	42
Placebo	1266 (44)	822 (30)	534 (16)	329 (7)	137 (1)	50

B Major Bleeding or Death from Any Cause



No. at Risk (no. of events)

Edoxaban	1270 (57)	866 (41)	551 (30)	324 (11)	145 (10)	44
Placebo	1266 (42)	829 (36)	538 (17)	332 (9)	138 (5)	49

Use the specified trial endpoint

AVERRROES - clinical AF unsuitable for warfarin

Aspirin / apixaban, % per year:

Stroke or systemic embolism: 3.7 / 1.6.

Major bleeding: 1.2 / 1.4.

ARTESIA - device-detected subclinical AF

Aspirin / apixaban, % per patient-year:

Stroke or systemic embolism: 1.24 / 0.78 (intention-to-treat).

Major bleeding: 0.94 / 1.71 (on-treatment).

Different populations and analysis sets: do not rank treatments across trials or use time-unspecified NNT/NNH. See Repair 11.

Wearables

- Photoplethysmography
- Single-lead ECG

→ Require EKG & clinical confirmation

- Limitations / Sources of Error
 - Motion
 - Skin contact
 - Discrimination b/w similar dysrhythmias
 - Longer episodes for detection when inactive

TABLE 2 Summary of the 3 Largest Mass Population-Based AF Screening Studies

	Huawei Heart Study (N = 187,912)	Apple Heart Study (N = 419,297)	Fitbit Heart Study (N = 455,699)
Algorithm accuracy			
Passive PPG PPV	91.6%	84%	98% PPV for the algorithm during ECG patch wear among 1,057 participants who had previously received an IHRD notification and then wore and returned an ECG patch monitor
User experience			
Active measurement	PPG recorded by bands or watches analyzed by smartphones	ECG recorded and analyzed by Apple Watch	PPG-based algorithm for detecting undiagnosed AF from a range of wrist-worn devices
Period measurement	Based on PPG Periodic measurement results given; the proportion of irregular rhythms analyzed	Based on PPG Periodic measurement results not given	Overlapping 5-min pulse windows (tachograms)
Period measurement frequency	Every 10 min	Every 2 h for baseline mode; irregular tachogram initiates a cascade of more frequent mode (every 16 mins)	Every 5 min
Closed-loop service			
Follow-up mode	Follow-up by mAFA platform, combined with clinical care AF management	Video visit with study doctor by app and mail ECG ePatch	Eligible participants with an IHRD, defined as 11 consecutive irregular tachograms, invited to schedule a telehealth visit and mailed a 1-week ambulatory ECG patch monitor
Diagnosis method for AF	Confirmed with medical history, physical examination, ECG, or 24-h Holter monitor by health care providers	By the results of ePatch	Ambulatory ECG patch monitor
Follow-up	62%	Only 44% of participants initiated contact with a telehealth doctor when prompted by the app, and among those receiving ePatches only 21% were returned and analyzed	Of participants who received IHRD notifications, 1,671 (35.3%) completed a telehealth visit
Percentage of confirmed AF by doctors or ePatch	87%	34%	32.2%

Key Messages: Prolonged Cardiac Rhythm Monitoring for Secondary Stroke Prevention

- AF detected after stroke is common but confers lower risk
- Prolonged monitoring increases AF detection, but unclear if it reduces recurrent stroke risk
- AF burden is a determinant of IS risk
- Burden of subclinical, device-detected AF to confer net clinical benefit w/ AC (reduction in stroke risk > hemorrhage risk) is not clear (but likely small)
- AF begets AF (AHREs → subclinical AF → clinical AF)
- Wearables will likely supplant current rhythm monitoring approaches

Left Atrial Appendage Occlusion (LAAO)

- 80-90% thrombi → LA appendage (NVAF)
 - TEE, surgery, autopsy
 - Cardioversion: TEE
 - Acute stroke: cardiac CT
- 50-60% in valvular (rheumatic) AF

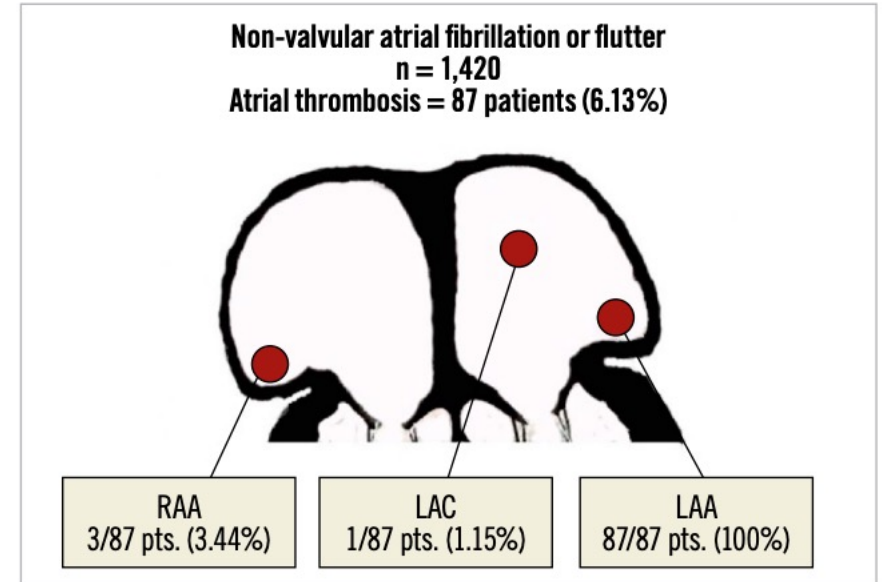
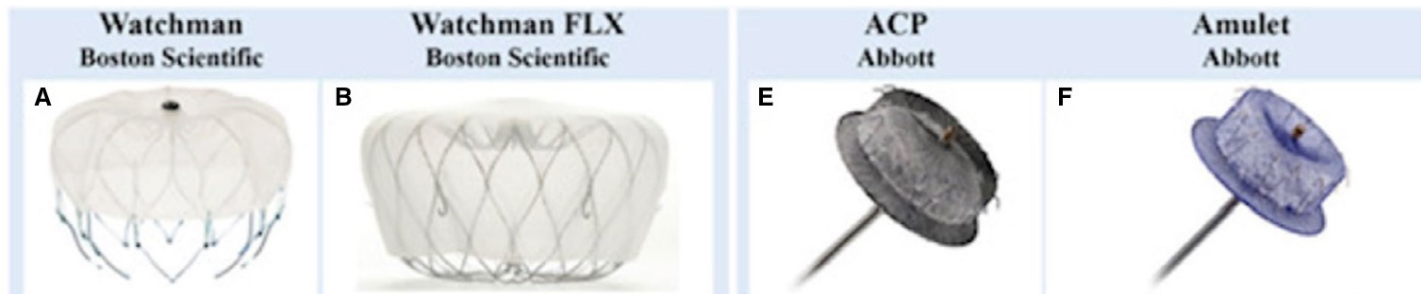
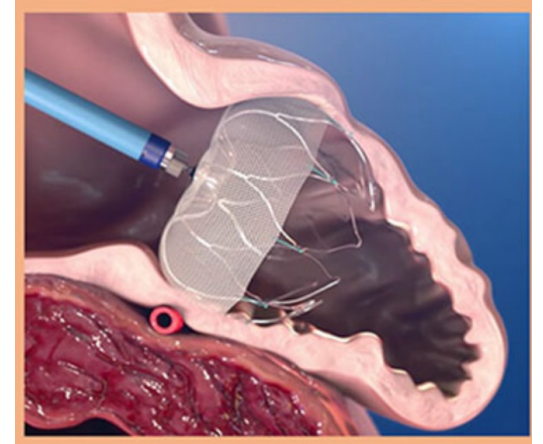


Figure 2. The localisation of atrial thrombi in the population study is represented in the schema.



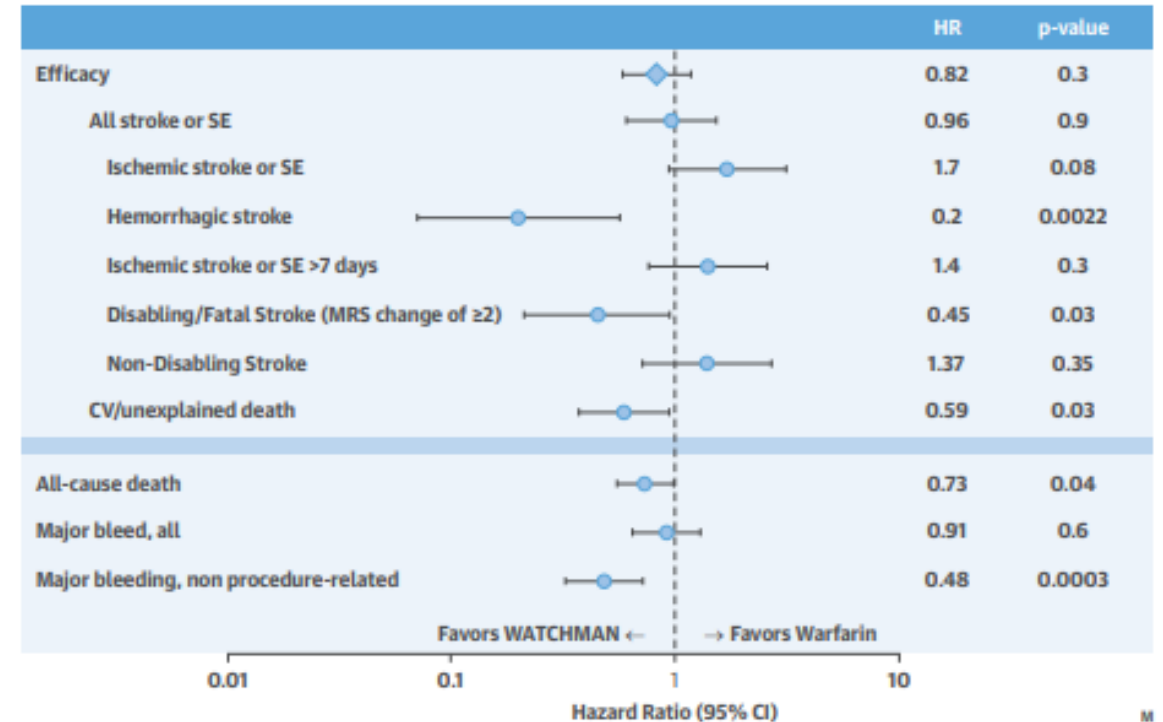
Ann Thorac Surg. 1996 Feb;61(2):755-9.
EuroIntervention. 2019 Jun 12;15(3):e225-e230.
J Am Heart Assoc. 2021 Nov 2;10(21):e022274.
Eur Stroke J. 2023 Mar;8(1):168-174.

Percutaneous LAAO & Warfarin



- PROTECT AF (N=707)
 - Non-inferiority, open-label
 - *Population*: nonvalvular AF eligible for AC, CHADS₂ ≥ 1
 - *Intervention/Control*: LAAO-Watchman / warfarin (2:1)

Historical: warfarin + aspirin initially; after imaging-guided cessation, DAPT until 6 months postimplant, then aspirin.
 - *Outcome*: composite (stroke, SE, CVD death)
- PREVAIL (N=407)
 - FDA mandated confirmatory study



Percutaneous LAAO & DOACs

- PRAGUE-17 (N=402)
 - Non-inferiority, open-label
 - *Population*: nonvalvular AF +
 - major hemorrhage, cardioembolic event i/s/o OAC, &/or CHADS₂VASc≥3 + HAS-BLED>2
 - *Intervention/Control*: LAAO-Watchman/Amulet or DOAC
 - DOAC → ~96% apixaban
 - *Outcome*: composite (stroke, TIA, SE, CVD death, hemorrhage, complications)

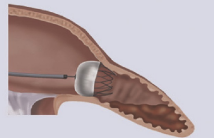
J Am Coll Cardiol. 2022 Jan 4;79(1):1-14.

CENTRAL ILLUSTRATION: A Summary Slide of Primary and Secondary Endpoints

PRAGUE-17 Trial: Long-Term (4-Year) Follow-Up

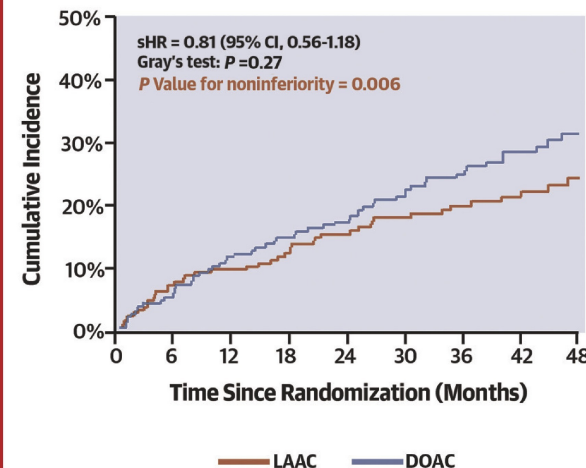


- 402 High-risk AF pts → Randomized
 - CHA₂DS₂-VASc = 4.7 ± 1.5
 - HAS-BLED = 3.1 ± 0.9
- Median Follow-up: 3.5 years (IQR 2.6-4.3), 1,354 pt-year

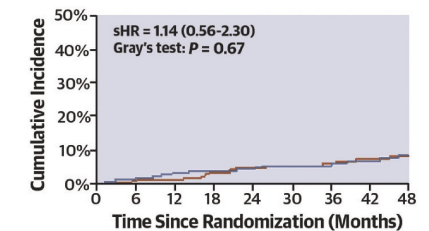


Primary Endpoint

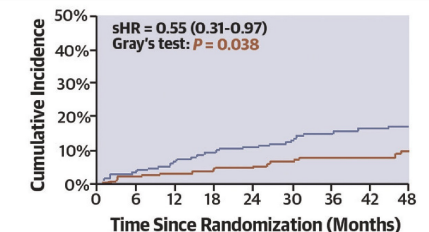
Stroke, TIA, SE, CV Death, Bleeding or Complications



Stroke or TIA

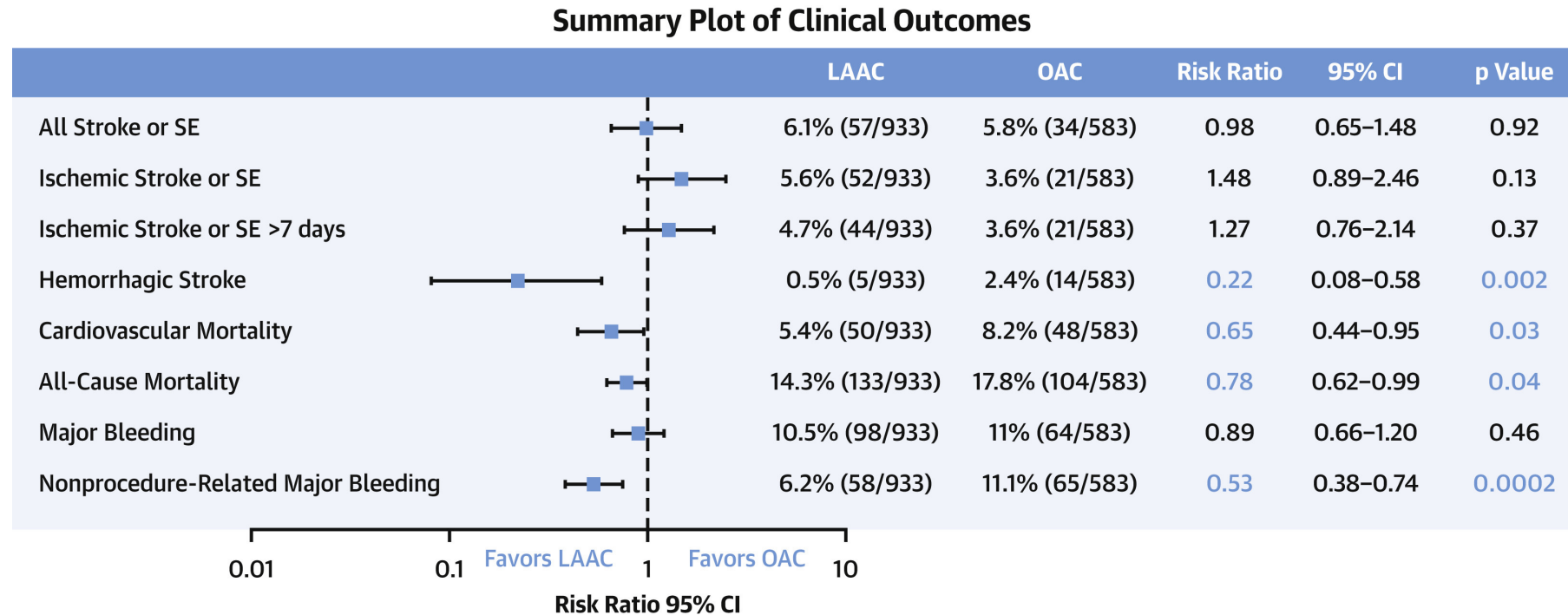


Non-Procedural Clinically Relevant Bleeding



Osmancik, P. et al. J Am Coll Cardiol. 2022;79(1):1-14.

Percutaneous LAAO



- Risks → device-related thrombus (2-5%), residual leak
- Post-procedural antithrombotics

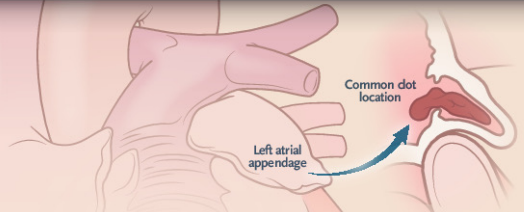
Historical FLX: OAC + aspirin initially; after 45-day TEE-guided OAC cessation, aspirin + P2Y12 inhibitor until 6 months postimplant, then aspirin. Current device IFU must be checked. See Repair 17.

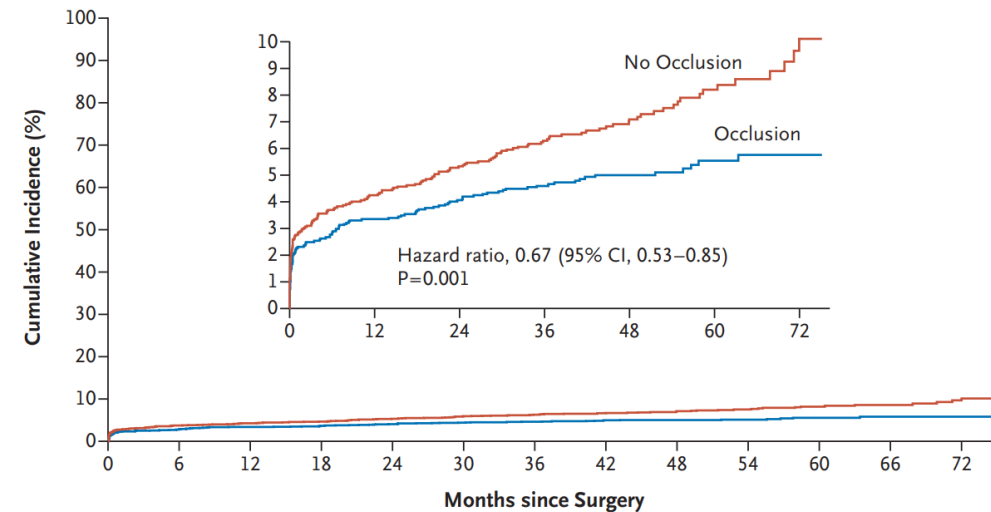
Surgical LAAO

- LAAOS-III (N=4811)
 - *Population*: AF + CHADS₂VASc ≥ 2 undergoing cardiac surgery
 - *Intervention/Control*: surgical LAAO / no LAAO
 - *Outcome*: Ischemic stroke / SE
- Mean CHADS₂VASc ~4
- ~77% OAC (3 y)

N Engl J Med. 2021 Jun 3;384(22):2081-2091.

Left Atrial Appendage Occlusion during Cardiac Surgery to Prevent Stroke

MULTICENTER, RANDOMIZED, CONTROLLED TRIAL		
 4811 Adults with a history of atrial fibrillation and a CHA₂DS₂-VASc score of ≥2 undergoing cardiac surgery	Cardiac surgery plus left atrial appendage occlusion	Cardiac surgery without left atrial appendage occlusion
	N=2400	N=2411
Ischemic stroke or noncerebral systemic embolism during follow-up	4.8%	7.0%
	HR, 0.67; 95% CI, 0.53–0.85; P=0.001	
Oral anticoagulation at 3 yr	75.3%	78.2%
Left atrial appendage occlusion during cardiac surgery reduced risk of ischemic stroke or systemic embolism in participants with a history of atrial fibrillation		



No. at Risk													
No Occlusion	2391	2134	2081	2030	1981	1897	1607	1291	1016	751	540	348	205
Occlusion	2379	2163	2105	2059	2020	1948	1642	1322	1046	781	550	349	199

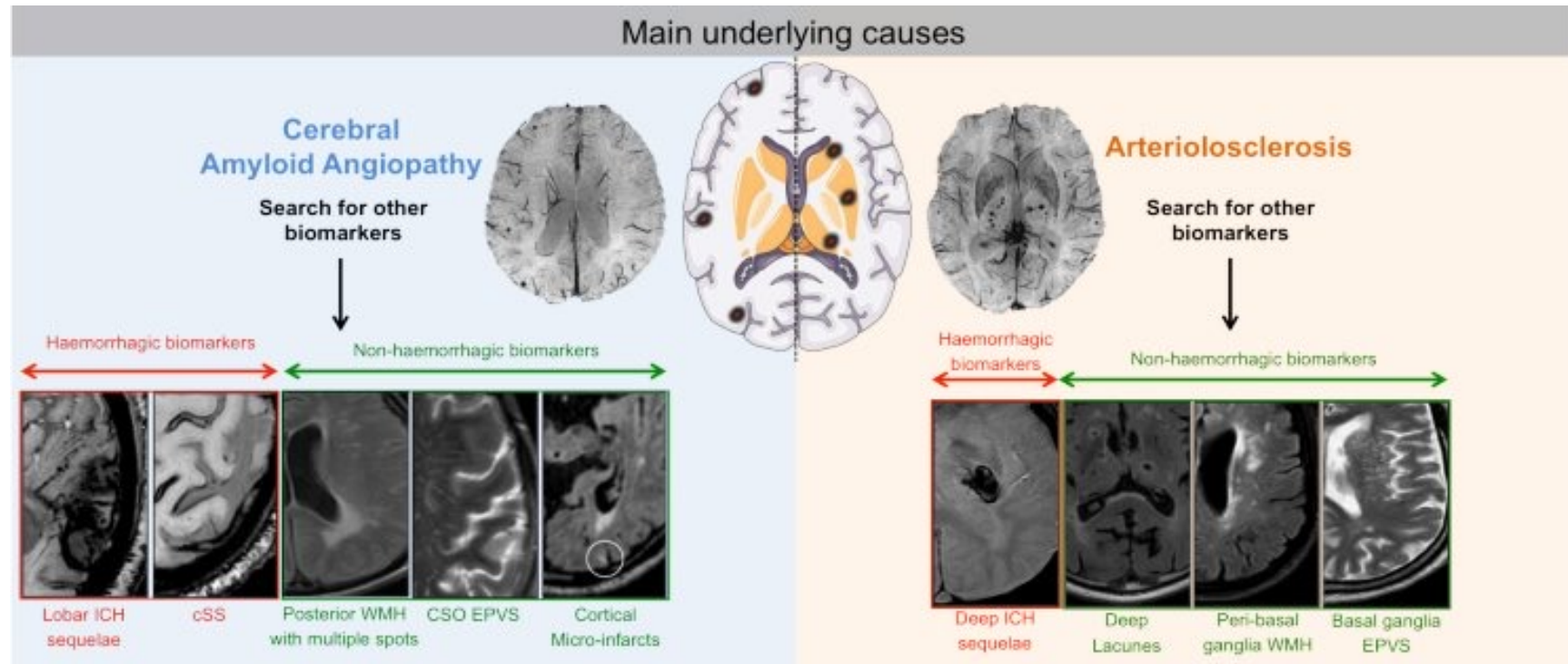
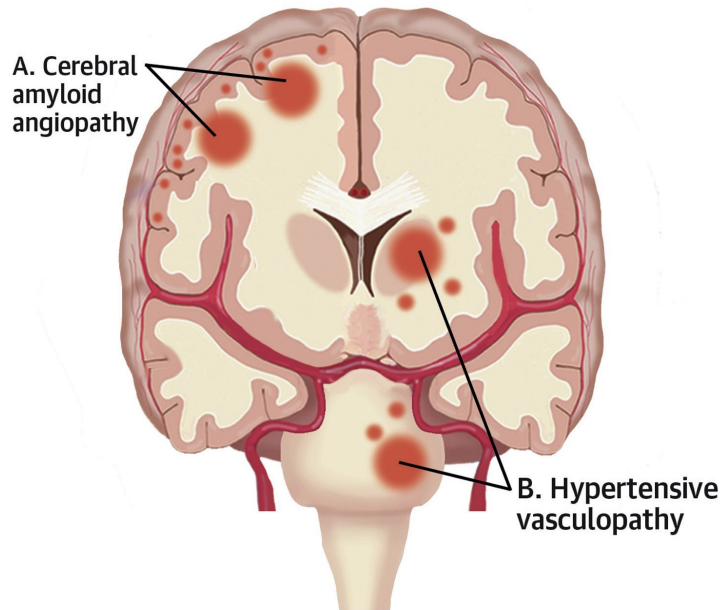
Figure 1. Cumulative Incidence of Stroke or Systemic Arterial Embolism.

	Left atrial appendage occlusion (LAAO)			
	PROTECT-AF (NCT00129545) ⁶⁸	PREVAIL (NCT01182441) ⁶⁹	PRAGUE-17 (NCT02426944) ^{70,71}	LAOOS-III (NCT01561651) ⁷²
Trial characteristics				
Sample size	707 participants	407 participants	404 participants	4770 participants
Intervention	Percutaneous LAAO	Percutaneous LAAO	Percutaneous LAAO	Surgical LAAO
Comparator	Warfarin	Warfarin	DOAC	No LAAO
Anticoagulation	Warfarin 50% (control group)	Warfarin 50% (control group)	DOAC 50% (control group)	82% (64% VKA, 18% DOAC)
History of ischaemic stroke	131 (18%) participants	111 (28%) participants	129 (35%) participants	9% of participants
Outcomes				
Follow-up	3.8 years, mean	1.5 years, mean	3.5 years, median	3.8 years, mean
Primary endpoint	Stroke, systemic embolism, and cardiovascular or unexplained death 8.4% (LAAO) vs 13.9% (warfarin), RR 0.60; 95% CI 0.41–1.05	Stroke, systemic embolism, and cardiovascular or unexplained death 0.064 (LAAO) vs 0.063 (warfarin), RR 1.07; 95% CI 0.57–1.89	Stroke, transient ischemic attack, systemic embolism, cardiovascular death, major or non-major clinically relevant bleeding, or complications related to procedure or device 10.99% (LAAO) vs 13.42% (NOAC), HR 0.84; 95% CI 0.53–1.31; p=0.44	Ischemic stroke or systemic embolism 4.8% (LAAO) vs 7.0% (OAC), HR 0.67; 95% CI 0.53–0.85
Stroke	1.4 (LAAO) vs 1.1 (warfarin) events per 100 patient-years	1.9% (LAAO) vs 0.7% (warfarin)	1.9% (LAAO) vs 1.8% (DOAC)	4.6% (LAAO) vs 6.9 (no LAAO)
Mortality	Cardiovascular or unexplained death 1.0 (LAAO) vs 2.4 per 100 years (warfarin)	Cardiovascular or unexplained death 2.6% (LAAO) vs 2.2% (warfarin)	All cause mortality 7.0% (LAAO) vs 7.8% (DOAC)	All cause mortality 22.6% (LAAO) vs 22.5% (no LAAO)

AF antithrombotic management after ICH

- (Annual) Recurrence Risk

- CAA: ~9%
- HTN: <2%
- Mixed: ~5%



J Am Coll Cardiol. 2020 Apr 21;75(15):1819-1831.

J Neurol Neurosurg Psychiatry. 2021 Feb 9;jnnp-2020-323951.

Neurology. 2018 Jan 9;90(2):e119-e126.

Aspirin

- RESTART (N=537)
 - *Population*: ICH taking antithrombotic
 - *Intervention/Control*: start / avoid AP
 - *Outcome*: recurrent sx ICH
- AF → ~25%
- AP start → median 76 d

J Neurol Sci. 2018 Jan 15;384:133-138.
Lancet. 2019 Jun 29;393(10191):2613-2623.

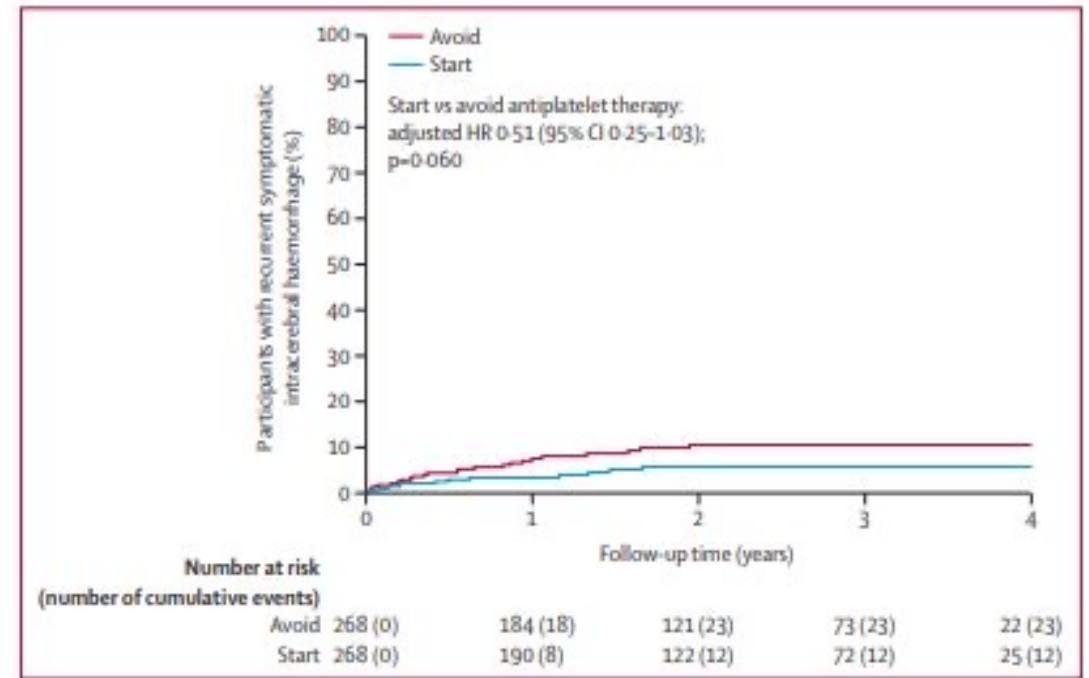
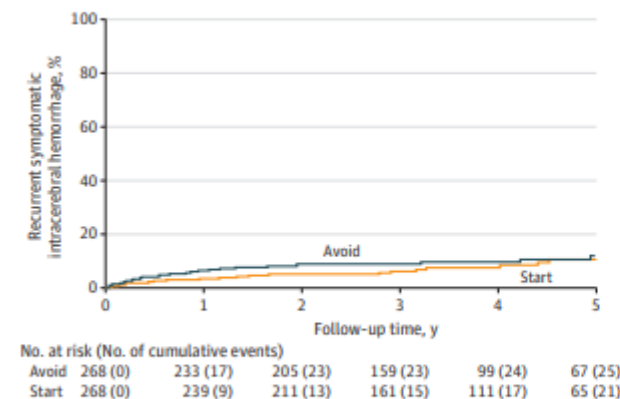


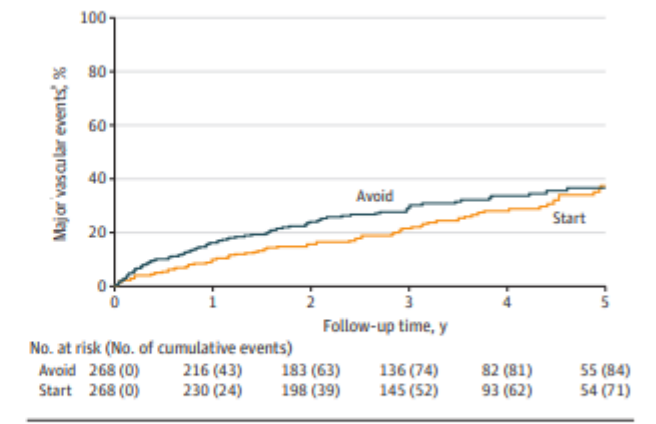
Figure 2: Kaplan-Meier plot of the first occurrence of recurrent symptomatic intracerebral haemorrhage

Figure 2. Risk of the First Occurrence of Recurrent Symptomatic Intracerebral Hemorrhage



Numbers at risk refer to survivors undergoing follow-up at the start of each year according to treatment allocation. Plot was censored at 5 years. Cumulative events indicate the participants in follow-up with a first event.

Figure 3. Risk of the First Occurrence of a Major Vascular Event



Numbers at risk refer to survivors under follow-up at the start of each year according to treatment allocation. Plot was censored at 5 years. Cumulative events indicate the participants in follow-up with a first event.

Anticoagulation

- CHADS2VASc > estimated risk of ICH recurrence?
- Uncertain benefit of AC resumption (stroke/CVD death)
 - Timing → ~8 weeks

Table 3. Ongoing Post-ICH Stroke Prevention Trials in Patients With Nonvalvular Atrial Fibrillation

Trial Name	Intervention vs comparator	Projected n	Country(ies)
ASPIRE	Apixaban vs aspirin	700	United States
ENRICH AF	Edoxaban vs aspirin	1200	Canada+20 countries
PRESTIGE AF	Anticoagulant vs antiplatelet vs none	654	United Kingdom, Austria, France, Italy, Germany, Spain

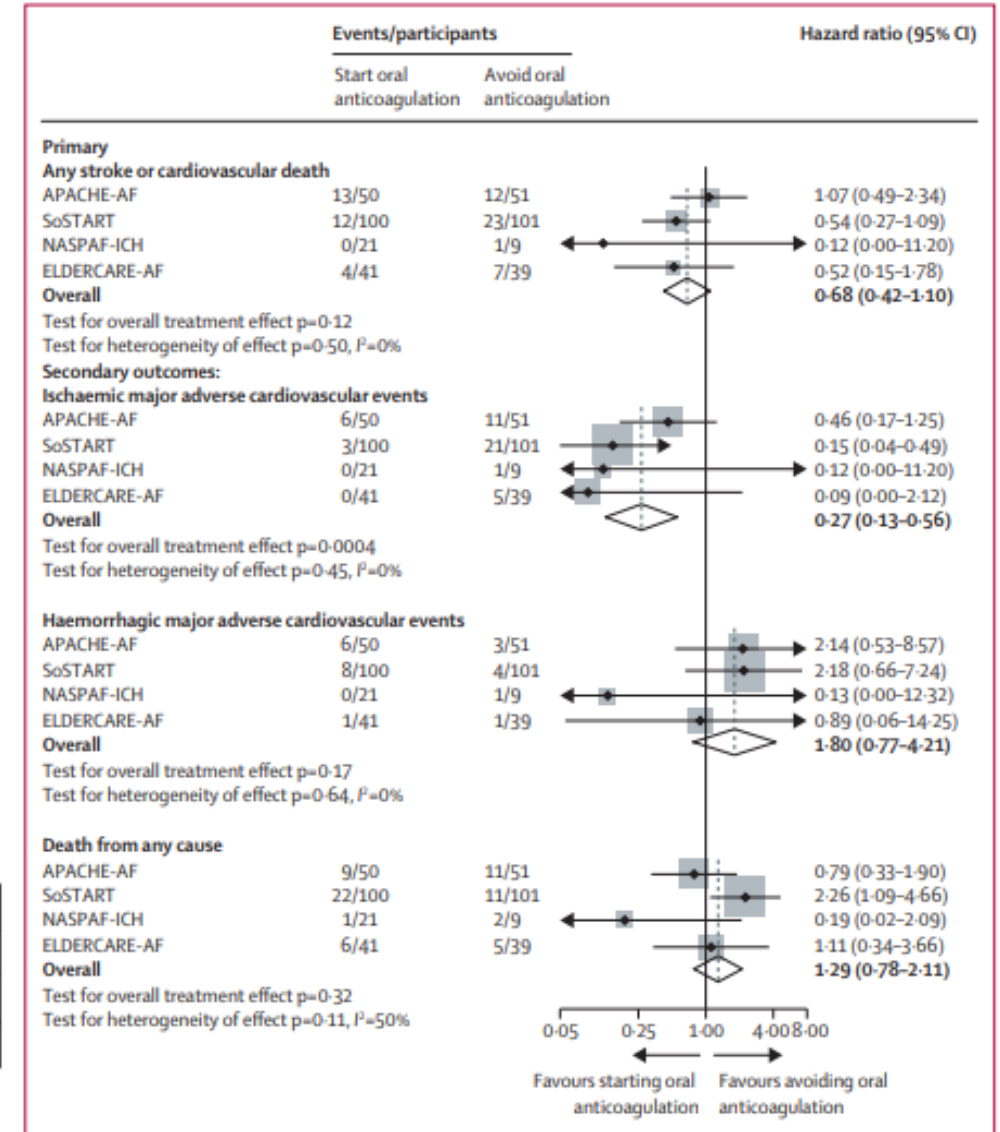


Figure 2: Effect of starting versus avoiding oral anticoagulation for atrial fibrillation after intracranial haemorrhage on primary and secondary outcomes, using individual participant data from four trials

Percutaneous LAAO

- Observational studies

- Retrospective
- N = small

→ Low event rates after LAAO

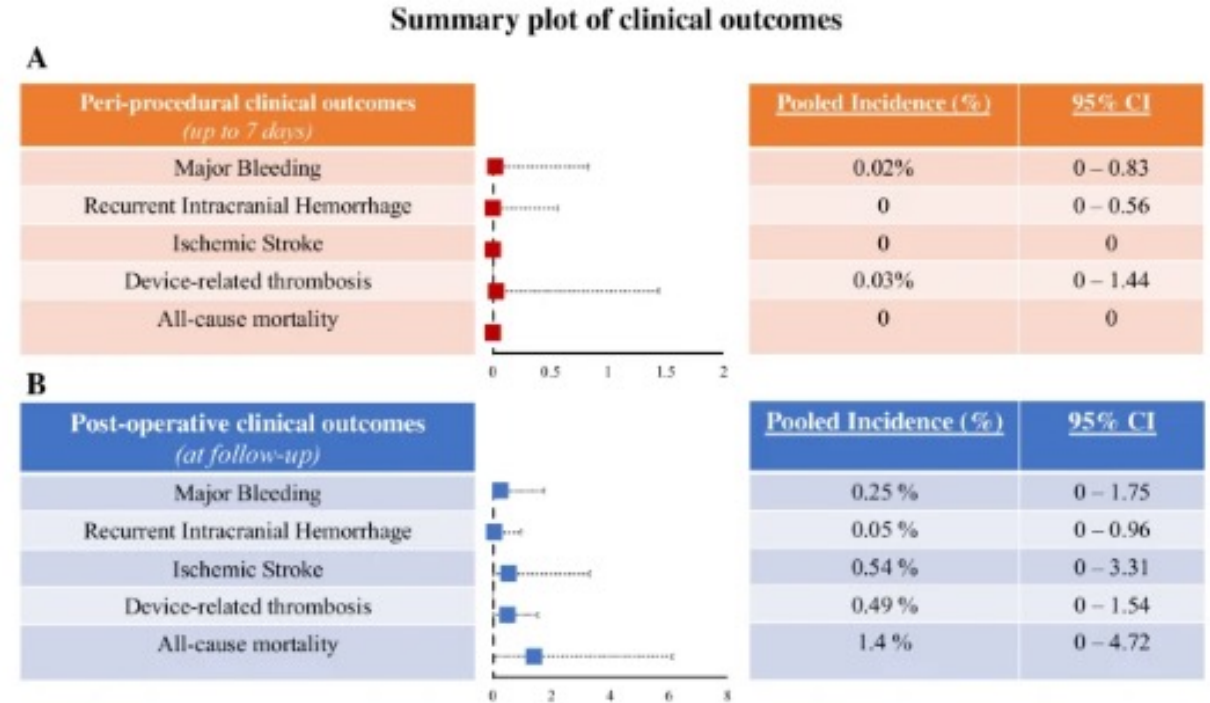


Table 3. Ongoing Post-ICH Stroke Prevention Trials in Patients With Nonvalvular Atrial Fibrillation

Trial Name	Intervention vs comparator	Projected n	Country(ies)
CLEARANCE trial	WATCHMAN FLX vs DOAC	550	Germany
STROKECLOSE	Amulet vs DOAC	750	Denmark
A ₃ ICH	DOAC vs LAAC vs ASA/none	300	France

EuroIntervention. 2017 Jun 20;13(3):371-378.

J Interv Card Electrophysiol. 2020 Nov;59(2):415-421.

Transl Stroke Res. 2021 Apr;12(2):259-265.

J Interv Card Electrophysiol. 2022 Sep;64(3):551-556.

Key Messages – LAAO & Antithrombotic Mgmt After ICH

- LAAO reasonable consideration when hemorrhage risk is unacceptably high, BUT:
 - Benefit : Risk (in comparison to DOAC) not established in stroke populations
 - LAAO + DOAC superior to DOAC alone in ischemic stroke despite AC?
- Optimal antithrombotic mgmt. after ICH in AF populations is uncertain
 - Enroll into ASPIRE whenever possible

RESTART studied selected prior-antithrombotic ICH survivors (median 76 days after ICH). It does not establish safety for every phenotype or start time, or make aspirin a substitute for AF anticoagulation.



Targeted repairs | 30 September 2026

The original deck is an archive. These numbered notes document targeted corrections; unchanged material has not been comprehensively updated. Trial selection, source conflicts and unread correction notices remain explicit. Original slide numbers are preserved.

Repair 02 - ELAN: study design and treatment windows

ELAN used an estimation design without a prespecified superiority or noninferiority hypothesis. Early DOAC: within 48 hours for minor/moderate stroke, days 6-7 for major stroke. Later: days 3-4 for minor, 6-7 for moderate, 12-14 for major. Overall admission NIHSS was median 5 (IQR 2-11); the randomization NIHSS median was 3. A universal enrollment cutoff of 100 hours is not established by the cited main report. The recovered PDF declares an update on June 29, 2023; the update substance was not independently reconciled.

ELAN | <https://pubmed.ncbi.nlm.nih.gov/37222476/>

ELANprotocol | <https://pmc.ncbi.nlm.nih.gov/articles/PMC9720853/>

ELANfull | <https://eprints.gla.ac.uk/299352/2/299352.pdf>

Repair 09 - Renal DOAC table: essential footnotes

The retained cells reproduce 2023 AF guideline Table 19, but must be read with its footnotes. Apixaban 2.5 mg twice daily applies when at least two of these are present: age at least 80 years, weight at most 60 kg, serum creatinine at least 1.5 mg/dL. ARISTOTLE excluded creatinine greater than 2.5 mg/dL or calculated CrCl less than 25 mL/min. Nonrenal considerations, including interactions, also apply. Gray cells identify doses not studied in pivotal trials. For rivaroxaban at CrCl less than 15, the guideline notes limited pharmacokinetic data and distinguishes the AF indication from other indications. This archived guideline table is not a stand-alone dosing instruction: current drug labeling and individual eligibility must be checked. The three indexed AF2023 correction bodies were reviewed separately; none changed these numbered recommendations or Table 19. The cropped slide does not reliably preserve evidence shading: consult the full source Table 19 rather than interpreting this image's gray rows as evidence categories.

AF2023 | <https://pmc.ncbi.nlm.nih.gov/articles/PMC11095842/>

Repair 10 - TIMING: randomization versus treatment window

TIMING randomized patients within 72 hours, comparing DOAC initiation within 4 days with initiation on days 5-10. Noninferiority was demonstrated using a 3 percentage-point margin; superiority was not demonstrated. Do not substitute the randomization window for the treatment window.

TIMING | <https://pmc.ncbi.nlm.nih.gov/articles/PMC9648987/>

Repair 11 - Cross-trial comparison: endpoints and analysis populations

Keep endpoints and analysis populations separate. AVERROES: stroke/systemic embolism was 3.7%/year with aspirin and 1.6%/year with apixaban; major bleeding was 1.2%/year and 1.4%/year, respectively. ARTESIA: primary stroke/systemic embolism was 1.24%/patient-year with aspirin and 0.78%/patient-year with apixaban (intention-to-treat); major bleeding was 0.94% and 1.71%/patient-year (on-treatment). The 1.02%/0.64% pair in ARTESIA was ischemic or unknown stroke, not its primary endpoint. The original table combined different populations/endpoints and time-unspecified NNT/NNH; it has been replaced without cross-trial efficacy ranking.

AVERROES | <https://pubmed.ncbi.nlm.nih.gov/21309657/>

ARTESIA | <https://pubmed.ncbi.nlm.nih.gov/37952132/>

ARTESIAfull | <https://openaccess.sgul.ac.uk/id/eprint/115973/1/nejmoa2310234.pdf>

Repair 12 - TAVI with AF: trial-specific benefit and bleeding

After TAVI with AF, ENVISAGE met noninferiority for its composite endpoint, but major bleeding was higher with edoxaban: 9.7 versus 7.0 events per 100 patient-years; HR 1.40 (95% CI 1.03-1.91). ATLANTIS did not demonstrate apixaban superiority; its anticoagulation-indicated stratum was exploratory, not an equivalence demonstration. The ATLANTIS erratum body remains unread, so correction-current clearance is not claimed.

ENVISAGE | <https://pubmed.ncbi.nlm.nih.gov/34449183/>

ATLANTIS | <https://pubmed.ncbi.nlm.nih.gov/35583186/>

Repair 13 - Valve disease: warfarin exception scope

The AF2023 warfarin exception specifies rheumatic mitral stenosis or mitral stenosis of moderate or greater severity, and mechanical valves. It does not apply to all rheumatic valve disease or all degrees of nonrheumatic mitral stenosis. Source: section 6.8.5, recommendations for AF in valvular heart disease.

AF2023 | <https://pmc.ncbi.nlm.nih.gov/articles/PMC11095842/>

Repair 14 - AZALEA: participants with prior stroke

The AZALEA secondary analysis Table 1 reports prior ischemic stroke in 191 of 1,287 participants (14.9%). A blanket prior-stroke exclusion is incorrect. This repair does not update all trial recruitment statuses elsewhere on the archived slide.

AZALEA | <https://pmc.ncbi.nlm.nih.gov/articles/PMC12781132/>

AZALEAregistry | <https://clinicaltrials.gov/study/NCT04755283>

Repair 15 - Stroke-risk scores: hypertension and calibration

CHA2DS2-VASc hypertension includes current antihypertensive treatment, including controlled treated hypertension; it is not limited to a currently elevated blood pressure. Source: AF2023 Table 10 risk-factor definition. Original absolute-risk bar values were not traced to a primary cohort; they have been withheld as individual risk estimates. Absolute risk varies with population, endpoint, follow-up and competing risks.

AF2023 | <https://pmc.ncbi.nlm.nih.gov/articles/PMC11095842/>

Repair 16 - RATE: brief episodes are not a treatment threshold

RATE found that brief device-detected episodes were not associated with clinical events. A bare 20-second label is not a validated stroke-risk or anticoagulation threshold. The exact recording-duration definition was not verified from the full paper; this repair is limited to the primary abstract conclusion.

RATE | <https://pubmed.ncbi.nlm.nih.gov/27754946/>

Repair 17 - WATCHMAN: historical antithrombotic sequence

Historical WATCHMAN FLX labeling specifies anticoagulant plus aspirin initially; after approximately 45-day TEE, anticoagulant cessation is conditional on device seal and thrombus assessment. Aspirin plus a P2Y12 inhibitor then continues until six months after implantation, followed by aspirin. Six months is measured from implantation, not from the 45-day visit. This restores the historical sequence, not a current universal device regimen; current device-specific instructions and alternatives were not reviewed. Sources: FDA P130013S035C, physical pages 59-60, and PINNACLE FLX methods.

FDAFLX | https://www.accessdata.fda.gov/cdrh_docs/pdf13/P130013S035C.pdf

PINNACLE | <https://pmc.ncbi.nlm.nih.gov/articles/PMC10111496/>

Repair 18 - RESTART: population and timing limits

RESTART enrolled selected survivors of ICH that occurred while using antithrombotics, randomized a median 76 days after ICH. Its results do not establish antiplatelet safety for every ICH phenotype or start time, and aspirin is not a substitute for AF anticoagulation. A universal aspirin-safety statement has been replaced with this population and timing qualification.

RESTART | <https://pmc.ncbi.nlm.nih.gov/articles/PMC6617509/>